NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR304069
Lead Applicant	Miss Florence Cook
Research Title	SUSTENANCE: Development of a toolkit to improve poSt-operative feeding practices and nUritional intake in patientS wiTh hEad and Neck cANCer undergoing surgEry with flap tissue reconstruction.
Plain English Summary	**Aim:** This study aims to improve the nutrition of patients with head and neck cancer when moving from tube-feeding to eating by mouth, after surgery. **Why is this research important?** In the UK, 12,400 people are diagnosed with head and neck cancer each year, of which 75% have surgery. This may involve removing parts of the mouth used for eating/drinking, such as the tongue. The surgeons will rebuild these by using tissue from another part of the body, such as the arm. This is called a 'flap'. After surgery, patients do not eat or drink for six to twelve days, and nutrition is provided by feeding tube either through the nose or directly into the stomach. The type of tube placed, the timeframe for re-starting eating/drinking and when the tube can be removed depends on many factors. These include being able to confidently swallow enough food/drink safely (affected by the amount of swelling in the mouth) and local hospital policies for when tubes are removed. Generally, the surgeon and team decide this for each person. However, nose tubes are often removed as soon as eating/drinking has begun, and before patients can manage enough food/drink. This happens because these tubes can prevent going home in some hospitals. Previous research shows 40-72% patients do not eat/drink enough after surgery. Nutrition drastically affects recovery and is the third largest cost-save to the NHS. Good nutrition speeds up recovery, improves wound healing, survival, and wellbeing. Poor nutrition slows recovery, increases risks of infection and even death. **How will I do this?** We want to understand what factors help or stop patients restart eating/drinking enough so that feeding tubes can be removed at the right time. Also, we think some patients can eat/drink sooner. This will improve their nutrition and help swallowing muscles recover. • We will first review research articles that have studied patients moving from tube-feeding to eating/drinking. • We will then survey healthcare staff including surgeons, nurses, dietitians, and speech therapists across sixty-five UK hospitals. We will find out how practices differ including when patients restart eating/drinking and what types of tubes patients go home with. • We will use the information from the survey and articles to decide what data we will collect from hospital records. We will also decide which staff practices we will watch and questions we need to ask patients and staff. When it comes to eating/drinking, the hospital data will tell us when patients restart, whether they have enough, and reasons for delayed or not enough eating/drinking. The questions and practices watched will help us understand how patients find eating/drinking after surgery and how staff help them. Findings will be used to develop a staff toolkit to improve feeding practices and nutrition after surgery. **How have patients been involved?** We discussed this project with patients who have had flap surgery. They told us that this research felt meaningful to them, and they lacked confidence with restarting eating/drinking as this often occurred abruptly and with little support. This made them feel isolated. We will work with patients across all stages of this project to: • Alert us to missed areas. • Check study information sheets are clear. • Check the data we plan to collect. • Check questions we will ask patients and staff to ensure these are clear/relevant. • Review patient interview logs to check we understand the patients' viewpoint. **How will findings be shared?**

	We will share with patients through social media, blogs, our funder's website, charities and support-groups by making an animated video of findings. We will share with staff working in head and neck cancer by writing articles and speaking at meetings.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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1. Application Summary Information

Host organisation (which will administer any award)
University College London Hospitals NHS Foundation Trust

Partner organisation	University College London

Research Title	SUSTENANCE: Development of a toolkit to improve poSt-operative feeding practices and nUritional intake in patientS wiTh hEad and Neck cANCer undergoing surgEry with flap tissue reconstruction.

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
1.00

Duration (months) The duration for your award is based on your chosen WTE
36

Proposed start date if grant awarded	01/09/2024

Research Type	Other (please specify)

Please provide more details on your selection above.
Mixed methods study

2. Applicant CV

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/2017	08/2019	MSc	Clinical Nutrition & Eating Disorders	United Kingdom (England)	University College London	Distinction
09/2009	08/2013	MNutr	Master of Nutrition (with state registration in Dietetics)	United Kingdom (England)	University of Nottingham, The	First Class Honours

Present and previous positions

(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
10/2023	-	Band 7 Senior Specialist Head & Neck Oncology Dietitian	University College London Hospital NHS Foundation Trust	0
03/2023	09/2023	Band 7 Senior Specialist Head & Neck Oncology Dietitian	University College London Hospital NHS Foundation Trust	40
12/2022	03/2023	Band 8a Clinical Lead Head & Neck Oncology Dietitian (Fixed-term secondment covering mat leave)	University College London Hospital NHS Foundation Trust	40
05/2022	11/2022	Band 8a Clinical Lead Head & Neck Oncology Dietitian (Fixed-term secondment covering mat leave)	University College London Hospital NHS Foundation Trust	20
12/2021	09/2022	Teaching Fellow in Nutrition & Dietetics (Fixed-term secondment) 2 days/week for 10 months	King's College London	0
06/2020	04/2022	Band 7 Senior Specialist Head & Neck Oncology Dietitian	University College London Hospital NHS Foundation Trust	0
06/2019	05/2020	Band 8a Clinical Lead Head & Neck Oncology Dietitian (Fixed-term secondement covering mat leave)	University College London Hospital NHS Foundation Trust	0
06/2016	06/2019	Band 7 Senior Specialist Head & Neck Oncology Dietitian	University College London Hospital NHS Foundation Trust	0
02/2015	05/2016	Band 6 Head & Neck Oncology Dietitian	University College London Hospital NHS Foundation Trust	0
11/2013	02/2015	Band 5 Rotational Dietitian (Elderly Care & Oncology/Haematology)	Heart of England NHS Foundation Trust, Birmingham	0
09/2013	11/2013	Research Assistant	University of Nottingham Medical School	100

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
UCLH Chief Nurse Research Fellowship	Principal applicant for a Chief Nurse Research Fellowship (£20,000) awarded in open-competition via the UCLH CNMAR (Centre for Nurse, Midwife and AHP-led Research). This provided me with 2 days/week for 10-months (December 2022 to September 2023) to develop a doctoral application, improve my publication outputs and undertake research methods training.	20000	University College London Hospital Charity	10	01/12/2022	29/09/2023
UCL/UCLH Biostatistics department fund	Principal applicant for a discounted place (£200 from £550) on the UCL Practical Medical Statistics course awarded in open-competition from UCL/UCLH Biostatistics department.	350	University College London	1	11/06/2023	16/06/2023
Patient and Public Involvement (PPI) starter grant	Principal applicant for a PPI starter-grant (£300) which was awarded in open-competition from the NIHR UCLH/UCL Biomedical Research Centre to inform this research proposal. Funds were used to cover subsistence costs to form a PPI group and conduct an initial group discussion.	300	UCLH/UCL NIHR Biomedical Research Centre	4	31/08/2022	13/01/2023
NIHR/HEE ICA Pre-doctoral Bridging Programme Place	Principal applicant for an NIHR/HEE Integrated Clinical Academic Pre-doctoral Bridging Programme Place (£7780) awarded in open-competition. This included £2780 to cover MRes module fees 'PhD Proposal Development' (King's College London) and £5000 to backfill my time 1 day/week for 6 months (May 2022 to November 2022).	7780	Health Education England/ NIHR	6	31/05/2022	30/11/2022
MSc studentship	Principal applicant for a UCL Studentship awarded in open-competition by UCL Division of Medicine for MSc Clinical Nutrition tuition fees (£12,680).	12680	University College London	24	24/09/2017	04/08/2019

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Cook, F., Howland, D, Wade-McBane, K, Freeman, L and Smith, R. (2023) ‘Nutritional management in the treatment of head and neck cancer: United Kingdom National Multidisciplinary Guidelines.’, *Journal of Laryngology*

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
and Otology, (Accepted manuscript).
Chartkul, M, **Cook, F.** and Govender, R. (2023) ‘Interpretation services used for non-English speaking patients with head and neck cancer : an audit from the joint dietetic and speech therapy pre-treatment clinics’, *Connect Journal*, 15(Accepted manuscript).
Armoogum, J, Taylor, V, Cruickshank, S, Donaldson, J, Tanay, M, Holdsworth, F, **Cook, F.**, McNamara, J, Job, C, Rees, T, Pope, L, Howell, C, Parfitt, L and Potter, L. (2023) ‘Development of online cancer resources to support pre-registration nurses and allied health professionals to increase their knowledge and understanding of cancer care.’, *British Journal of Nursing*, 32(17).
Cook, F., Glen, K and Ngo, AT. (2023) ‘Service improvement for a dietetic-led radiologically inserted gastrostomy service using the Plan-Do-Study-Act cycle.’, *Clinical Nutrition Open Science*, (51: 1-10).
Cook, F., Nazareth, I. and Govender, R. (2023) ‘Developing skills of allied health professions for a principal investigator role: A case from the SIP SMART2 swallowing prehabilitation trial.’, *Advances in Communication and Swallowing*, (26(1):53-61).
Cook, F., Govender, R and Brennan, PA. (2023) ‘Greetings from your predatory journal! What they are, why they are a problem and how to spot and avoid them.’, *British Journal of Oral and Maxillofacial Surgery*, (61(3):245-247).
Cook, F., Rodriguez, J and McCaul, L. (2022) ‘Malnutrition, nutrition support and dietary intervention: the role of the dietitian supporting patients with head and neck cancer.’, *British dental journal*, (233:9).
Cook, F., Ubhi, J, Fong, K, Sinha, D and Kalavrezos, N. (2022) ‘Oral feeding commencement following free or pedicle flap surgery for oral cancer: experience from a tertiary referral centre.’, *Oral Oncology Reports*, (3-4).
Cook, F., Langdon-Daly, J and Serpell, L. (2022) ‘Compliance of participants undergoing a 5-2 intermittent fasting diet and impact on body weight.’, *Clinical nutrition ESPEN*, (52:257-261).
Schache, A., Kerawala, C., Ahmed, O., Brennan, **Cook, F.**, Garret, M, Homer, J, Hughs, C, Mayland, C, Mihai, R, Newbold, K., O'Hara, J., Roe, J., Sibtain, A., Smith, M., Thavaraj, S., Winter, L, Young, V and Winter, S. (2021) ‘British Association of Head and Neck Oncologists (BAHNO) Standards 2020.’, *Journal of oral pathology & medicine : official publication of the International Association of Oral Pathologists and the American Academy of Oral Pathology*, (100:1).
Monen, A, **Cook, F.**, Harrison, H and Kainth, P. (2020) ‘Retrospective review of nutritional status in head and neck oncology patients during radiotherapy treatment.’, *Journal of human nutrition and dietetics : the official journal of the British Dietetic Association*, (33 (Suppl. 1), 35–41).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/12/2022	British Dental Journal	Article included in the 2022 British Dental Journal 'Big Impact Collection' (top accessed/cited/shared): Cook et al (2022) Malnutrition, nutrition support and dietary intervention.	First author
01/11/2022	National Institute for Health and Care Research	Accredited Associate Principal Investigator for the SIP SMART 2 swallowing prehabilitation trial in head and neck cancer. 6-month tenure (May 2022 to November 2022).	Associate Principal Investigator
01/12/2019	British Dietetic Association Research	'Best Abstract Prize' in the Nutrition Support stream for	Students MSc dissertation supervisor

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
	Symposium	'Retrospective review of nutritional status in head and neck oncology patients during radiotherapy treatment'.	
01/11/2019	University College London	Dean's List Commendee for highest overall performing student in MSc programme. Commends outstanding academic performance, equivalent to the top 5% of student achievement in the faculty.	Awarded commendation
01/07/2013	University of Nottingham	Norma Lauder Prize for highest mark achieved in final year module 'Advanced Dietetic Practice' during undergraduate MNutr degree.	Awarded prize

ORCID iD 0000-0001-7405-6080

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Dietitian

Do you have current / active professional registration? This registration must be with one of the **ICA approved regulatory bodies**
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
DT27179 - HCPC

Please describe your research career to date

I am a dietitian and developing clinical-academic who strives to improve the nutritional care of patients with head and neck cancer (HNC) through applying evidence-based nutritional interventions into practice. Witnessing the positive impact nutrition has on physical/psychosocial wellbeing motivates me to pursue an independent role in research and contribute new knowledge to improve/innovate nutritional care.

Research experience

My enthusiasm for research started during my undergraduate dissertation which investigated the diets of university athletes. I gained ethical approval, recruited six participants, collected dietary data, and explored dietary choices through a focus group. I developed skills in statistical (using SPSS) and thematic analysis.

Throughout my career, I have developed research skills in quantitative data analysis, dissemination (journal publications and conference presentations), and stakeholder engagement through conducting 10 service evaluations and audits to resolve problems encountered in day-to-day practice. The satisfaction gained from seeing direct improvements in care makes me appreciate pursuing research within a clinical role. For example, following concerns that the local protocol for gastrostomy insertion caused unnecessary placement delays, I audited complications, benchmarked pathways, and formed a stakeholder group to agree policy changes. A re-audit revealed reduced delays without increased complications, published in *Clinical Nutrition-OpenScience*.

My interest in enhanced recovery after surgery (ERAS) began with a service evaluation revealing patients wait for prolonged periods before recommencing oral intake post-surgery. I published findings in *Oral Oncology Reports* and presented at the British Association of Head and Neck Oncologists (BAHNO) conference. I realised that evidence-gaps were best answered through research and could change perioperative nutritional care. The NIHR Clinical-Academic pathway presented the ideal opportunity to provide training/resources towards doctorate and beyond, steering me to successfully apply for an NIHR-Health Education England (HEE) pre-doctoral bridging award (November 2021). This important milestone included an MRes module in *PhD-Proposal Development* and helped me develop my ideas through academic supervision and patient feedback, after I was awarded a PPI starter-grant. I was then awarded a UCLH Chief-Nurse Research Fellowship (June 2022), to develop a doctoral application, undertake statistics training, and improve my publication outputs.

As a developing clinical-academic, I have taken opportunities to acquire research skills whilst in practice. I am an NIHR-accredited Associate-PI on SIPSMART2 prehabilitation trial. Alongside learning all aspects of clinical trial delivery, I successfully led recruitment at UCLH to target, recruiting 33% of eligible participants. I showcased my research skills in *Advances in Communication and Swallowing,* demonstrating development opportunities across all Vitae-RDF domains. I demonstrated capability of allied health professionals in leading trial delivery with fellow Associate-PIs, at the BAHNO conference. I was invited to share my experience at the NIHR conference: *Research, the NHS and you*; improving awareness of clinical research opportunities accessible in practice. Furthermore, when a studentship for MSc Clinical Nutrition tuition-fees was advertised, I successfully applied and

completed alongside full-time clinical work which included taught modules in '*Research Methods'.* I developed academic writing skills through my dissertation, which was a quasi-experimental study assessing participant adherence with intermittent-fasting diets, published in *Clinical Nutrition-ESPEN*. I was awarded the UCL Dean's List Commendation in recognition of outstanding academic achievement.

Collaboration and professional engagement

I have developed national and international networks though election to the BAHNO-Council (representing dietetics) and BAHNO-International Collaboration Committee. I have helped organise four annual BAHNO conferences to-date by marking abstracts and chairing sessions.

I co-authored national standards published in *Journal of Oral Pathology*; the benchmark document for quality assurance in HNC used nationally and abroad. I led ENT-UK national guidance for dietetics; the key document that guides practitioners on optimal service provision/delivery, used in NHS business cases to fund dietetic services (accepted in *Journal of Laryngology;* first-author). I worked with Restorative Dentistry UK, contributing a dietetics article for the *British Dental Journal* HNC-edition. This provided guidance on mitigating risks of oral nutritional supplements on oral health and made the 2022 *British Dental Journal* 'Big Impact Collection'.

I am gaining recognition in cancer nutrition, receiving invitations to:

- Peer-review manuscripts for journals including *Supportive Care Cancer* and university healthcare ethics applications, developing my critical-appraisal skills.
- Speak at London Sarcoma and UCLH-Support in Cancer Research Forum, developing my presentation skills.
- Review resources for Sarcoma-UK and World Cancer Research Fund, developing my public-engagement skills.
- Join the British Dietetic Association (BDA)-HNC Committee.
- Lecture on UCL nutrition/dietetics degrees.

Supervisory and teaching

I completed a 10-month secondment as a Teaching Fellow at Kings College London, developing my educational and leadership skills. I co-led, redesigned, and revalidated a 30-credit dietetics module. I was personal tutor to five students, providing pastoral care.

I represent dietetics on HEE Aspirant Cancer-Career and Education-Development programme (workstream-1), reviewing e-learning modules in cancer-care, which increased knowledge and understanding amongst aspirant cancer professionals (published in *British Journal of Nursing;* co-author).

I have successfully supervised 6 MSc and 2 BSc dissertations to completion. One student audited weight change during radiotherapy. This raised awareness of weight loss risks amongst patients with obesity. We published the abstract in *Journal of Human Nutrition and Dietetics* and were awarded the 'Best Abstract Prize' at the BDA Research Symposium.

Professional experience

I have developed specialist skills in HNC through advanced communication training and MSc Cancer-Care modules (Royal Marsden School). I have extensive clinical supervision experience, having developed staff-training competencies in HNC. I was the department's student-training co-lead 2019-2020 and helped set-up the departments research group to develop a research culture in dietetics.

I have developed leadership and management skills through seconding as Clinical-Lead HNC Dietitian twice (over 2-years). My responsibilities included team management, complex workforce planning amid COVID-19, staff recruitment/retention, developing business cases and utilising research funding within a clinical service. I have contributed to coroners' inquests, instilling in me the importance of evidence when justifying clinical decisions.

I am applying for this fellowship as though I have made great progress as a developing clinical-academic, this is the crucial next step in becoming a clinical research leader. My ambitions include leading a research programme in perioperative nutrition at UCL, supporting UCLH become an ERAS Centre of Excellence and nurturing dietetic talent to produce a sustainable team of clinical academics.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
No

Contextual Factors
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding, or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

4. Plain English Summary of Research

Aim:

This study aims to improve the nutrition of patients with head and neck cancer when moving from tube-feeding to eating by mouth, after surgery.

Why is this research important?

In the UK, 12,400 people are diagnosed with head and neck cancer each year, of which 75% have surgery. This may involve removing parts of the mouth used for eating/drinking, such as the tongue. The surgeons will rebuild these by using tissue from another part of the body, such as the arm. This is called a 'flap'.

After surgery, patients do not eat or drink for six to twelve days, and nutrition is provided by feeding tube either through the nose or directly into the stomach. The type of tube placed, the timeframe for re-starting eating/drinking and when the tube can be removed depends on many factors. These include being able to confidently swallow enough food/drink safely (affected by the amount of swelling in the mouth) and local hospital policies for when tubes are removed. Generally, the surgeon and team decide this for each person. However, nose tubes are often removed as soon as eating/drinking has begun, and before patients can manage enough food/drink. This happens because these tubes can prevent going home in some hospitals. Previous research shows 40-72% patients do not eat/drink enough after surgery. Nutrition drastically affects recovery and is the third largest cost-save to the NHS. Good nutrition speeds up recovery, improves wound healing, survival, and wellbeing. Poor nutrition slows recovery, increases risks of infection and even death.

How will I do this?

We want to understand what factors help or stop patients restart eating/drinking enough so that feeding tubes can be removed at the right time. Also, we think some patients can eat/drink sooner. This will improve their nutrition and help swallowing muscles recover.

- We will first review research articles that have studied patients moving from tube-feeding to eating/drinking.
- We will then survey healthcare staff including surgeons, nurses, dietitians, and speech therapists across sixty-five UK hospitals. We will find out how practices differ including when patients restart eating/drinking and what types of tubes patients go home with.
- We will use the information from the survey and articles to decide what data we will collect from hospital records. We will also decide which staff practices we will watch and questions we need to ask patients and staff.

When it comes to eating/drinking, the hospital data will tell us when patients restart, whether they have enough, and reasons for delayed or not enough eating/drinking. The questions and practices watched will help us understand how patients find eating/drinking after surgery and how staff help them.

Findings will be used to develop a staff toolkit to improve feeding practices and nutrition after surgery.

How have patients been involved?

We discussed this project with patients who have had flap surgery. They told us that this research felt meaningful to them, and they lacked confidence with restarting eating/drinking as this often occurred abruptly and with little support. This made them feel isolated. We will work with patients across all stages of this project to:

- Alert us to missed areas.
- Check study information sheets are clear.
- Check the data we plan to collect.
- Check questions we will ask patients and staff to ensure these are clear/relevant.
- Review patient interview logs to check we understand the patients' viewpoint.

How will findings be shared?

We will share with patients through social media, blogs, our funder's website, charities and support-groups by making an animated video of findings. We will share with staff working in head and neck cancer by writing articles and speaking at meetings.

5. Scientific Abstract

Background

Surgery is a widely accepted treatment for head and neck cancer (HNC). After surgery, patients are made 'nil-by-mouth' for approximately 6-12 days to enable wound healing and nutrition is provided via a feeding tube placed through the nose (naso-gastric tube) or directly into the stomach (gastrostomy). Variation exists for time to first oral intake, type of feeding tube placed and when these are removed, with limited evidence to determine optimal practice. Many patients experience delays in recommencing oral intake which deconditions swallowing musculature. During the transition from tube to oral intake, 40-72% of patients have inadequate nutrition due to premature removal of feeding tubes. Optimising nutrition is critical in supporting post-operative recovery and improves wound healing, survival, and quality-of-life.

Research question

Development of a toolkit to improve post-operative feeding practices and nutritional intake in patients with HNC undergoing surgery with flap tissue reconstruction.

Aims and objectives

This research aims to investigate post-operative feeding practices for patients with HNC undergoing surgery when transitioning from tube to oral intake and to develop a staff toolkit to improve timing/adequacy of intake. The nature of the toolkit will be determined by conducting exploratory work as part of evidence synthesis. This will be achieved by examining the literature, describing current practice, investigating nutritional outcome data, exploring patient experiences, staff perceptions and practices.

Methods

A mixed-methods convergent-parallel approach over three years, across five workstreams:

1. Scoping review of existing literature to map current post-operative feeding practices and identify knowledge gaps necessitating further inquiry.
2. Cross-sectional web-based survey of healthcare professionals working in UK HNC centres (n=65) to map variation in practices and describe current reality including typical time to first oral intake, type of tube placed and home tube-feeding status.
3. Prospective observational cohort study (n=90) to investigate nutritional status and intakes during the transition from tube to oral intake, time/type of first oral intake and associated factors. Multiple regression analysis to identify predictors of nutritional adequacy.
4. Qualitative approaches using semi-structured interviews and observation exploring patient experiences (n=10) and healthcare professionals (n=16-18) perceptions and practices of transitioning from tube-feeding to optimal oral intake. Framework Analysis of transcripts and fieldnotes.
5. Development of a post-operative feeding toolkit with stakeholder consultation.

Impact

Patients:

Improved nutritional outcomes by identifying facilitators and barriers for achieving nutritional adequacy and optimal timing for removing feeding tubes to avoid nutritional deficits. Improved return to swallowing function by identifying optimal timing for re-commencing oral feeding.

Healthcare professionals:

Potential to change perioperative HNC feeding management by developing a toolkit that optimises decision-making when transitioning patients from tube to oral intake, accounting for variation in local resources and infrastructure.

NHS/service:

It is anticipated that streamlining oral feeding commencement and avoiding post-operative nutritional deficits will reduce hospital stays, readmissions and associated costs.

Dissemination

Results will be disseminated in peer-reviewed journals targeted to the HNC readership and through conferences, professional networks and websites including the NIHR and charities. To reach the wider audience, an animation will be shared via social-media, blogs and support groups according to patient and public involvement advice.

6. Detailed Research Plan

What is the problem?

In the UK, 1 in 2 people will have cancer in their lifetime. Disease burden is increasing with more people living longer with cancer[1]. The NHS long-term plan identified improving cancer care as a national priority[2]. Head and neck cancer (HNC) is the 8th most common cancer in the UK and 4th in men. Incidence has grown by a third in the last decade[3], associated with rises in human-papilloma virus related HNC[4]. 12,422 people are diagnosed annually of which 75% require surgery[3]. Surgery involves removal of important structures used for swallowing which are then reconstructed using tissue from another part of the body, called a 'flap'. This is known as 'flap-tissue reconstruction'. Following surgery, patients are placed nil-by-mouth (no eating/drinking) to allow surgical healing. Nutrition is delivered via a feeding tube through the nose (naso-gastric tube) or directly into the stomach (gastrostomy) until they can recommence oral intake. Current practices are demonstrated in Figure 1 and have two key problems:

Firstly, recommencing oral intake is likely unnecessarily delayed for most patients. At UCLH, patients wait on average 11-days[5], this could be reduced to 5-days for some patients[6]. Patients recommence oral intake when the surgeon is satisfied the site has recovered and the patient can swallow safely, often with input from the speech-therapist. If not achieved, this could lead to breakdown of the reconstruction, resulting in complications including wound breakdown, oroctuaneous fistula (passage between oral cavity and skin) or aspiration (passage of food into the lungs)[7]. Many surgeons conservatively prolong nil-by-mouth, even if the site has healed as there is limited evidence to determine when oral intake can be recommenced safely. However, delaying oral intake deconditions swallowing muscles and could be the difference between eating/drinking again or requiring life-long tube-feeding.

Secondly, patients often experience nutritional deficits when transitioning from tube to oral intake which can lead to developing malnutrition. This occurs because naso-gastric tubes get removed once oral feeding commences, irrespective of whether a patient is meeting their nutritional requirements orally[8], as naso-gastric tubes can prevent/delay discharge[9]. Some patients continue with tube-feeding at home but are generally those needing to remain nil-by-mouth, rather than those who can eat/drink but aren't meeting their nutritional requirements.

This proposal aims to improve this pathway by filling evidence-gaps in post-operative feeding practices, to avoid delayed and inadequate intake when transitioning from tube to oral feeding. This aligns with the 2019 James-Lind Alliance *Nutritional Screening and Malnutrition* research priority 'can nutritional-interventions lead to health improvements'[10].

Why is this research important?

This research falls under enhanced recovery after surgery (ERAS) initiatives. It should optimise post-operative feeding practices and nutrition[11]with patient, service, and economic benefits:

Patients benefits:

- Improved nutritional status by avoiding post-operative nutritional deficit; a modifiable risk factor for malnutrition which is negatively associated with muscle-mass, immune response, wound healing, fatigue, complications, morbidity/mortality, length-of-stay (LoS) and return-to-work[12].
- Improved swallowing rehabilitation by reducing inactivity time which deconditions swallow musculature[13].This could be the difference between rehabilitating swallow or requiring life-long tube-feeding[8,14].
- Improved quality-of-life due to improved independence with eating/drinking[15].
- Timely recommencement of eating/drinking improves perception of recovery[8]. *'Food is medicine'* resonates with patients, family and carers, associating eating with recovery[16].

Service and economic benefits:

- Malnutrition is prevalent in HNC, affecting 60% at baseline. 80% lose weight during treatment[17,18].Surgery increases the risk of weight loss. Preventing malnutrition is the third-largest potential NHS cost-saving, estimated at £23.5 billion; >15% of the UK health and social care budget[19].Malnourished patients are much more prone to infection/illness, admitted to hospital more frequently and experience longer stays.
- Approximately 1 in 10 (8.8-19.4%) of HNC patients are readmitted within 30-days of flap-surgery. A study in the USA indicated this costs $9,070-$15,916/patient[20,21].Post-operative gastrostomy accounts for 71.4% of planned readmissions[22].Unplanned readmissions are mainly due to wound breakdown however, 9% are for nutrition and 2.2-15.4% for dehydration[21,23,24].
- Avoiding delays in recommencing oral feeding, or 'early oral feeding' should accelerate recovery and

reduce LoS[25,26]with its associated NHS costs[27].

This research should prevent both developing and/or exacerbating malnutrition through optimising post-operative nutrition, and decrease readmissions associated with nutrition/hydration but also wound healing, as this is directly associated with poor nutrition.

Review of existing evidence

Early oral feeding is well established in treatment of other cancer types[28] due to benefits including reduced LoS and improved post-operative nutritional adequacy with calls for high-quality research in HNC[29]. Five retrospective studies[5,7,9,30,31]and one pilot randomised controlled trial (RCT)[32]have been conducted in HNC flap-surgery with another RCT registered (NCT04787939). These are limited by inconsistent definitions of early oral feeding (which a registered scoping review aims to determine[33]) and selection bias[6]. Firstly, only patients without a tracheostomy were randomised to early oral feeding. Tracheostomy-use varies between centres. In centres that treat early-stage disease, tracheostomy-use is 13%[9]. In centres that treat more advanced-stage disease, tracheostomy-use is 88%[31].Advanced-stage disease makes up ~59% of cases[34]. Secondly, patients with previous radiotherapy (~30% of cases) were excluded[30,35].

Current trials exclude most patients having flap-surgery meaning findings have limited uptake. To avoid excluding patients, safety evidence is required on perceived risk factors for oral feeding commencement. Two retrospective studies have started to establish these[31,36].However, naso-gastric tube presence on discharge was used as a binary outcome measure for oral feeding resumption. These are not mutually exclusive as tube and oral feeding can occur concurrently. Also, not all hospitals permit discharge with nasogastric tubes which the proposed survey will map. Neither study recorded availability of suitable hospital meals or patient-centred factors, such as confidence with eating/drinking. Type of diets initiated were described as 'fluids, puree and normal' which are not validated descriptors. Staff attitudes/beliefs towards early oral feeding have yet been explored and are a potential barrier to change[9], given surgeons report favouring their trainer's practice over RCTs in other cancer types advocating early oral feeding[37–39].

Consensus guidelines for nutrition in surgery recommend continuing tube-feeding in patients unable to meet their requirements orally[27]. Recommendations are based on RCTs in gastrointestinal and orthopaedic surgery which found that continued tube-feeding prevents weight loss, reduces malnutrition and improves treatment tolerance and quality-of-life compared to oral intake only[40]. In HNC, one retrospective study[8] has assessed whether patients meet their nutritional requirements for 10-days post-surgery. 40% had inadequate intake, defined as meeting 60% of estimated nutritional requirements. A 75% cut-off is considered more robust[27] and when applied, has found 75% do not meet requirements[41]. No data was collected on protein intake. The early feeding RCT[32] also measured nutritional adequacy for 3-days post-operatively. Findings lack accuracy as adequacy was based on nutritional requirements calculated from average weights of 50-70-year-olds in China, rather than actual participant weights. Furthermore, the trial was run in 2022 but used weight data from 2014, which won't capture demographic changes, as more patients with HNC now present overweight/obese[42];who are more likely to have inadequate intakes post-operatively[8]. Neither study provided information on whether patients with inadequate intakes continued tube-feeding or follow-up information on how this may have impacted on nutritional status.

Existing studies are limited by selection bias, follow-up period, non-validated measures, and have missed factors reported by patients to be important. These gaps will be remedied by the proposed prospective cohort study. This will run in a centralised HNC centre which will include patients with advanced-stage disease, previous radiotherapy, and tracheostomy. Nutritional status will be measured at baseline and 30-days post-surgery using the PG-SGA (patient-generated subjective global assessment), which is considered gold-standard in HNC. A 30-day follow-up can capture clinically significant changes in nutritional status. Factors previously missed will be recorded including pre/post-operative swallowing function, hospital meal availability, and type of diet (using validated IDDSI-International Dysphagia Diet Standardisation Initiative descriptiors[43]). Nutritional requirements will be estimated using best-practice equations calculated from actual weights. Energy and protein intake will be measured using the 75% cut-off for adequacy and information recorded on whether patients with inadequate intakes continued with tube-feeding. The proposed qualitative studies will capture patient-centred factors and staff attitudes/behaviours by drawing on theories that identify influences on health professional behaviour to address barriers for implementing change.

What is the research question/aims and objectives?

Research question:

What are the post-operative feeding practices and nutritional intake of patients with HNC transitioning from tube to oral intake after undergoing flap-surgery, and how can practices and intake be optimised?

Aim:

To investigate post-operative feeding practices and nutritional intake in patients with HNC transitioning from tube to oral feeding after undergoing flap-surgery and to develop a toolkit for timely and adequate intake. The nature of the toolkit will be determined by conducting exploratory work (WS1-4) as part of evidence-synthesis.

Objectives:

1. To examine post-operative feeding practices and nutritional intake when transitioning patients from tube to oral feeding following flap-surgery (WS1).
2. To describe national post-operative feeding practices on when patients commence oral feeding and how they are transitioned off tube-feeding following flap-surgery (WS2).
3. To examine whether patients meet their nutritional requirements when transitioning from tube to oral feeding and identify factors predictive of nutritional adequacy and oral feeding commencement (WS3).
4. To explore patient experiences and perceived facilitators and barriers for timely and adequate oral intake following flap-surgery (WS4).
5. To explore staff perceptions and practices for transitioning patients from tube to oral feeding following flap-surgery (WS4).
6. To develop a toolkit with stakeholder consultation to optimise post-operative feeding practices and nutritional intake (WS5).

Project Plan

Underpinning theory

This study is based on the Medical Research Council (MRC) guidance on complex interventions, specifically focussing on phase 1, intervention development[44]. This will draw on normalisation process theory (NPT)[45]to understand the organisational structures that influence implementation of complex interventions and the theoretical domains framework (TDF)[46]to understand human behaviour in complex clinical environments to improve implementation.

WS1: Scoping review [Months 1-7]

Rationale:

There are currently no reviews published or registered investigating post-operative feeding practices and nutritional intake after HNC flap-surgery. A systematic review registered (CRD42022380436) aims to assess whether early oral feeding influences wound healing, swallowing-function, and LoS. Nutritional outcomes are not being assessed. A scoping review will map practices and nutritional intake, identify knowledge-gaps and compliment findings from the registered reviews.

Methods:

Scoping review ineligible for PROSPERO and therefore it will be registered with Open Science Framework or Figshare.

Arksey and O'Malley (2005) 6-step framework[47]:

1. *Research question:* What are the post-operative feeding practices and nutritional intake of patients with HNC transitioning from tube to oral feeding after undergoing flap-surgery?
2. Data extraction: Systematic literature search conducted with no date limits.
3. Study selection: Initial screening of all titles/abstracts and then full articles by myself and a second investigator independently against criteria:
   Inclusion criteria:

- Adults>18 years
- HNC flap-surgery
- Focus on practices, timing/adequacy when transitioning from tube to oral feeding.
- English language
- Original research, case-reports, conference-abstracts, guidelines and grey-literature

Exclusion criteria:

- Animal studies
- Children/adolescents
- Non-english

Cohen's-K statistic will be calculated to determine inter-rater agreement with discordant results reviewed

by a third investigator.
4. Charting: Study characteristics will be extracted and compiled for coding/validation using *Covidence* software.
5. Reporting: Results will be reported using PRISMA-Scr and mapped according to sources/types of evidence and limitations reported e.g., any relevant sources omitted.
6. Consultation/dissemination: Findings will be discussed with PPI to capture insights not identified. Findings will inform cohort study measures, survey and interview questions.

WS2: National Survey [Months 3-7]

Rationale:

Variation exists for timing to oral intake and processes for transitioning off tube-feeding[5,9,14,31]. This is associated with local resources for placing gastrostomies and policies for discharging with naso-gastric tubes. A survey will describe current reality and canvass opinions to understand the contexts the toolkit will be implemented.

Design:

A cross-sectional open web-based survey circulated within the UK.

Survey questions will be developed using the review findings, alongside discussions between the research team which includes key stakeholders (dietitian, speech-therapist, surgeon) and patients. Content validity will be established using Lawshe's method[48],by asking independent healthcare professionals to rate how essential each question is to determine which to include/exclude.

An online, self-completed semi-structured questionnaire will be developed using Qualtrics software and piloted with healthcare professionals reflective of the sample frame to ensure functionality. This will include multiple-option answer, five-fold Likert scales and free-text comments asking about the centre demographic, timing to oral feeding, tubes placed and removal processes.

Responses will be mandatory and options of 'not applicable' or 'rather not say' provided to ensure data-completeness, with an opportunity to review/change answers before submitting.

Sample:

Multi-disciplinary leads at UK/NHS HNC surgical units (n=65) will be identified from the National HNC Audit list[49]and approached via email with an explanation of the study and asked to disseminate within their units to staff involved in transitioning patients from tube to oral feeding after flap-surgery. A 1-month timeframe will be provided to complete the survey. Follow-up emails sent on a maximum 3 occasions if no response is received. The survey will also be disseminated through professional groups.

Analysis:

View, response, participation, and completion rates will be calculated, duplicate responses removed, and incomplete questionnaires considered for analysis pending data-completeness.

Quantitative data will be analysed using descriptive statistics and free-text comments analysed using content analysis.

Quality approach:

CHERRIES-Checklist for Reporting Results of Internet E-Surveys reporting-guidelines will be adhered to[50].

Approaching all units will capture variation and ensure representation of different centres (e.g., teaching, district-general hospital) and geographic location[51].

Questionnaire items will be alternated to prevent bias and adaptive questioning utilised to reduce the complexity and number of questions.

Outcome:

Findings will inform the cohort study measures, qualitative interview questions and enable purposive sampling for staff interviews to capture variation in practices, to ensure the toolkit developed is implementable in different units, and sites for a future multi-centre validation study.

WS3: Prospective cohort study [Months 10-29]

Rationale:

Limited empirical research has identified factors associated with timely/adequate nutritional intake after surgery. A prospective cohort study will identify factors that may be predictive of nutritional adequacy that can be targeted in the toolkit.

Research question:

What are the factors that contribute to timely and adequate oral intake in patients with HNC transitioning from tube to oral feeding after undergoing flap-surgery?

Methodology:

An observational single-centre exploratory prospective cohort study will be undertaken at UCLH which is a centralised London service for four NHS-trusts, covering a wide-geographic area representative of people from diverse, multiple ethnic backgrounds affected by HNC[3,52,53].UCLH offers all discharge pathways for oral/tube feeding which optimises generalisability, within the limitations of a single-centre design.

Data will be collected from electronic hospital records. All measures are routinely gathered except for the following patient-recorded measures:

- 1-week diet diary once oral feeding commences.
- PG-SGA questionnaire and weight are routinely collected pre-surgery, but not always post-surgery. Patients will therefore be asked to complete these 30-days after surgery which can be obtained virtually if required[54].

Electronic-scales and pre-paid envelopes will be provided to return the PG-SGA and diaries.

Primary outcome measure:

Difference in PG-SGA score from surgery to 30-days post-surgery is anticipated. PG-SGA is gold-standard for assessing nutritional status in HNC, comprising a patient-reported questionnaire and dietetic assessment[55]. Higher scores indicate higher risk/presence of malnutrition. PG-SGA is numerical which will also maximise the information that can be established and modelling potential as opposed to a binary measure.

Secondary outcome measures:

- Post-operative nutritional intake including energy (kcal) and protein (g) relative to estimated nutritional requirements using best-practice equations. 75% cut-off for nutritional adequacy.
- Time until first oral intake (days).
- Pre- and 30-day post-operative weight (kg), height (cm), BMI (kg/m^2) and weight-change (%).

Explanatory variables:

- Demographics: age, sex, ethnicity, diagnosis, relevant medical history e.g., previous chemo/radiotherapy, diabetes, alcohol/smoking history, performance-status.
- Nutritional interventions: oral nutritional supplements, feeding tube device, post-operative tube-feeding duration and proportion this contributed to intake, indications for continuing tube-feeding on discharge, whether patients self-managed or required support from family/carers/district nurses.
- Pre/post-operative swallowing-function, diets prescribed (IDSSI) and availability of hospital meals.
- Surgical characteristics: reconstructive site/flap, defect-volume (cm^3), neck-dissection, tracheostomy presence/weaning.
- Postoperative LoS.
- Complications (Clavien-Dindo classification) and re-admissions within 30-days of surgery.

Sample:

All patients meeting the inclusion criteria over 18-months will be identified from the MDT meeting and approached, which is felt to be a realistic timeframe within the overall study period.

Inclusion:

- Adults>18 years
- HNC oral-cavity diagnosis

- Curative flap-surgery

Exclusion:

- <18 years
- Unable to give informed-consent.
- Non-cancer
- Palliative surgery or laryngectomy.
- Contraindication to oral/tube feeding post-surgery.

Local UCLH audit data revealed that ~100 flap surgeries are performed annually. Anticipating 60% recruitment will result in a sample of 90. A pragmatic approach will be applied to building the prediction model using the 'rule of 10' calculation[56]. This deduces that you can only fit a model as many as n/10 parameters (where n=90), which corresponds to 9 coefficients that can be included into the model.

Analysis:

Diet diaries will be analysed using Nutritics nutritional analysis software and information from hospital catering.

Statistical analysis will be conducted using SPSS and continuous variables assessed for normality; presented as mean ± standard deviation for symmetrical data or median [interquartile range]. Categorical variables expressed using frequency and percentages. Repeated measures (PG-SGA score, weight, BMI and % weight change at baseline and 30-days) analysed using paired t-test or Wilcoxon signed-rank test pending data distribution.

For the primary outcome (factors associated changes in PG-SGA score), a multiple regression model will be used with backwards elimination via two-stage procedure. Each potential predictor will be considered on its own as a univariable model to ascertain which variables offer the most and least predictive ability. A multivariable model will then be built using the most predictive variables and the lasso procedure considered if there are too many predictor variables to quantify/alleviate overfitting the model. Multiple imputation for missing data conducted where suitable.

For secondary outcome measures, multiple logistic regression analysis will be utilised to calculate the odds ratio for not reaching 75% of estimated nutritional requirements when transitioning from tube to oral intake and backwards elimination to identify variables associated with not meeting requirements. To identify variables associated with time to first oral intake, either multiple or cox regression will be used depending on whether there is any censoring.

Quality approach:

Participants will be provided with the same weighing scale and instructions for taking measurements (e.g., time of day) to promote consistency. Diet diaries will be analysed by one dietitian (myself) for consistency.

STROBE-Strengthening the Reporting of Observational studies in Epidemiology reporting-guidelines will be adhered to[57]. Careful consideration to the sample size constraints will be discussed with a statistician when building the multivariable model. Log-transformation of data will likely be required to satisfy statistical assumptions.

Recommendations from the COMET (Core-Outcome Measures in Effectiveness Trials) initiative[58] have been deferred to when selecting outcome measures. Eight HNC core-outcomes are recommended which mainly pertain to survival and late-effects but also include swallowing-function[59].Measurements for swallowing will align with COMET-recommendations and discussed with Dr Govender.

Outcome:

Findings will identify predictors of timely/adequate oral intake for targeting in the toolkit and help inform most appropriate primary outcome measure for future studies.

<u>WS4: Qualitative enquiries [Months 10-24]</u>

Rationale:

Patient feedback reported it was unclear who was responsible for restarting oral intake and removing feeding tubes as inconsistent information was liaised by staff with scope for improved communication. Qualitative interviews will capture patient experiences and staff perceptions of the barriers and facilitators towards timely/adequate oral intake. Observations will help understand the setting in which the toolkit will be used.

Research question:

What are the facilitators and barriers for transitioning from tube to adequate oral intake in patients with HNC after undergoing flap-surgery?

Methodology:

Rapid qualitative research techniques will be utilised to optimise efficiency by running data collection and analysis in parallel. 1:1 semi-structured interviews will be conducted with patients and healthcare professionals and unstructured observations of staff practices.

Sample:

Patient interviews:

A subset of patients participating in the cohort study will be invited off-treatment within 6-weeks of surgery. Patients who have not recommenced oral intake in this timeframe will be excluded. Purposive sampling will be used to ensure maximal variation in terms of sex, age, ethnicity, disease site/stage, type of flap and feeding tube. General guidance indicates that 6-12 interviews is usually acceptable with little new information beyond 15[60]. Recruitment will be until data saturation is achieved. Based on a similar study in colorectal cancer, a sample of 10 is anticipated[61].

Staff interviews:

Healthcare professionals will be invited to participate using purposive sampling to represent different professional roles and years of experience. Sampling will be supported by survey data to capture variation in practices. Recruitment will be until data saturation is achieved. Based on a study of similar design, a sample of 16-18 healthcare professionals is anticipated[38].

Inclusion:
- >18 years
- Involved in transitioning patients from tube to oral intake after flap-surgery in a UK/NHS HNC post-operative setting (e.g. surgical-ward, critical-care)

Exclusion:
- Not involved in facilitating re-commencement of oral diet/fluids after flap-surgery in a UK/NHS HNC setting
- Short-term staff e.g., agency

Observations:

Unstructured non-participant observation of surgical ward rounds, multi-disciplinary inpatient, and outpatient reviews at UCLH. UCLH is a centralised HNC centre for four hospitals and represents all post-operative feeding practices.

An initial purposive sample (inclusion/exclusion criteria per interviews) of three patient cases (naso-gastric tube, pre- and post-operative gastrostomy) to capture different practices for transitioning patients from tube to oral feeding. Observation will be until data saturation is achieved.

Data collection:

Interviews:

1:1 semi-structured interviews will be conducted face-to-face to last ~60minutes, audio-recorded and sound checked. Interview notes will also be documented throughout. Video option offered via MS-Teams for staff interviews to enable including staff geographically distant to London and/or for convenience.

Interview guide questions will be exploratory, open-ended and informed by WS1-2 and PPI with prompts/probes to allow participants expand on answers. Staff interview guide will be underpinned by the TDF[46].

Observations:

Fieldnotes will be collected using guidance from Emerson and colleagues[62].Research objectives will be deferred to prior to observations, as a reminder of what observation aim to establish.

Jottings/headnotes will be made at the time of observation of initial impressions, unexpected events, reactions that indicate significant actions/events, how routine actions take place, and emerging patterns. Detailed fieldnotes will

then be transcribed depicting the scene.

Analysis:

Interviews will be transcribed using a professional transcription service to aid time-management and reduce error. All recordings will be listened to for data-credibility. NVivo software will be used to manage/code data and retrieve systematically.

After each interview and observation, notes and main findings will be entered into a Rapid Assessment Procedure (RAP) sheet, developed and agreed with Professor Vindrola-Padros. The RAP sheet will be piloted and amended during initial data collection and used to identify salient topics from which the transcripts can be deferred to, acting as a triangulation tool[63].

Rapid data analysis will be undertaken by combining the RAP sheet with Framework analysis[64,65]. Framework analysis has been selected due to its compatibility with rapid analysis and use in healthcare-related research. Framework identification for patient interviews will be discussed/agreed with Professor Vindrola-Padros. A sample of transcripts and fieldnotes will be analysed by Professor Vindrola-Padros and reviewed by my PPI group. Themes will be member checked.

Quality approach:

Yardley's criteria will be employed to ensure credibility, reliability and trustiness[66]. Observation will be limited to 3-hour timeframes, symbols/abbreviations used in jottings to optimise recall and non-verbal communication recorded. Detailed fieldnotes will be written immediately after leaving the site and discussion avoided with others to avoid psychological intrusion. Rapid methods can generate similar findings to conventional methods but require experienced researchers[67],which Professor Vindrola-Padros will support with.

A research journal will be maintained to record reflexive notes after interviews and observations. This will be acknowledged and reflected upon during analysis in terms of how this may have affected interpretation and how my presence could have changed fieldsite dynamics.

Outcome:

Findings will identify patient- and staff-reported facilitators and barriers for timely/adequate oral intake to inform the toolkit.

<u>WS5: Development of post-operative feeding toolkit [Months 25-31]</u>

Rationale:

Previous studies indicate the need for improved nutrition post-surgery[68].Patients have specifically requested further information on nutrition when HNC education interventions have been evaluated[69]. A staff toolkit will be developed to improve post-operative feeding practices and nutritional intake. This may include staff education/training on nutrition, practice recommendations on when to recommence oral intake, and prescription of post-operative nutritional supplements which have been well-received in colorectal cancer[70].

Methodology:

A convergence-coding matrix will be used to integrate findings from WS3 and WS4 to determine areas of agreement, silence, and dissonance[71](Figure 2). Desk-based development of toolkit will then be undertaken using WS1-4 and convergence findings, MRC logic-modelling and PPI feedback/consultation. Stakeholder consultation will then be conducted with six staff-representatives involved in decision-making (Consultant and Registrar Surgeon, Dietitian, Speech-Therapist, ERAS-Nurse, Ward-Nurse) and 2 PPI-members. Questions from the 16-item NPT toolkit will be used to frame the consultation and generate discussion to refine the toolkit which will be amended accordingly using a series of iterations.

Analysis and outcome:

The toolkit will be evaluated in post-doctoral work.

<u>Dissemination, Outputs and Impact</u>

<u>Dissemination:</u>

To enter society, health/social care:

- Seven articles (review, survey, cohort study, 2xqualitative studies, toolkit-development, co-produced paper) published in open-access in peer-reviewed journals targeted to HNC/ERAS readership.
- Presenting locally at UCLH-Support in Cancer Research Forum
- Presenting national/internationally through conferences targeted to HNC, nutrition and ERAS clinicians:
    - BAHNO: to present the review.
    - ERAS-World Congress: to present the survey.
    - British Association of Oral/Maxillofacial Surgeons: to present the cohort study.
    - BDA-Research Symposium: to present the qualitative studies and build networks for delivering a future multi-centre study.
- Podcasts for cancer-care e.g., RadChat
- Brief for the NIHR-webpage
- Sharing through professional networks e.g., BDA-HNC Group and social-media e.g., Twitter

To inform/engage patients and the public, an engaging audio-visual animation of findings will be developed and disseminated through social-media, blogs, support-groups and charities. I will also disseminate at *The Swallows* charity patient conference including the impact of PPI in this research.

Outputs and further funding:

This research will produce an evidence-based toolkit to improve post-operative feeding practices and nutritional intake. A PhD thesis will be deposited in the UCL repository.

Post-doctoral funding will be sought (NIHR DSE/ACAF) to answer the question: *'Does use of a post-operative feeding toolkit lead to improved nutritional outcomes for patients with HNC undergoing flap-surgery?'.* This will include a multi-centre validation study, feasibility, pilot and definitive trial comparing the toolkit to standard care.

Impact of outputs and barriers:

For clinical stakeholders:

This research could change perioperative HNC feeding management by introducing a novel, evidence-based toolkit. Barriers to adoption include shifting conservative practices and engaging external centres to support a multi-centre study. I have considered these issues through:

- Collaboration with international experts experienced in implementing novel ERAS interventions.
- Collaboration with HNC organisations to incorporate findings into national guidance/standards.
- Building HNC networks in my training/dissemination plan and selecting a practice supervisor with well-established surgical networks to support future multi-centre work.
- Collaboration with our UCLH ERAS-nurse for local implementation.
- Underpinning this work with implementation science theories/frameworks.

For the NHS/patients:

At the end of this PhD, this research should improve nutritional outcomes and safety evidence for transitioning from tube to oral feeding post-surgery. Local toolkit uptake is anticipated within five-years and nationwide within ten, pending an RCT demonstrating effectiveness. Policy impact in anticipated through recommending the toolkit in National Institute for Health and Care Excellence and ENT-UK clinical guidelines.

Cost-saving is anticipated by avoiding malnutrition and 9% (unplanned) and 72% (planned) readmissions caused by nutrition. Readmission costs from flap-surgery have been evaluated in the USA ($9,070-$15,916/patient). Most comparable UK estimates comes from ear/nose/throat surgery (£1052/patient)[72]. However, average readmission duration was only 1.3-days (UK) compared to 8-days (USA) which may explain the lower cost. Streamlining oral feeding commencement should reduce LoS by 3-4days/patient. NHS bed-costs are ~£300-500/day/patient[73], saving £90,000-200,000/year (UCLH) or £5850,000-13,000,000/year (all 65 UK-units).

Project management

My proposal has 5 workstreams over 36-months. My pre-doctoral fellowship and part-time teaching secondment have provided excellent foundations for managing multiple commitments to time. My GANTT chart demonstrates how courses and professional practice will be managed alongside. Generous timeframes have been allocated for items that could impact on deliverables.

My academic supervisors are experienced in supervising PhD students and NIHR fellows. We have agreed monthly joint-meetings. Professor Brennan will review studies from a surgical perspective and help with accessing the HNC community for recruitment and dissemination. I will meet my Thesis Committee 5x during my PhD and submit 6-monthly progress reports to ResearchFish.

Financial management: Reviewed quarterly with UCLH/UCL Joint-Research Office finance team.

Ethics

WS2 will require university ethics and approval sought in the first 3-months. WS3/4 will require HRA ethical approval and sought within the first 10-months.

Participation will be voluntary with written informed-consent. Survey participants will be asked to tick whether they are happy to be contacted for an interview and provide an email address. Purposive sampling will be used for interviews/observations to ensure inclusiveness/diversity. Patient interviews will involve talking about experiences following surgery. This will be discussed with PPI and considerations made e.g., distress protocol. A pragmatic consent process will be taken for observations by approaching participants at the time for verbal consent, and 1-2 days after for written consent given dynamics of post-operative care. If they do not consent, fieldnotes will be securely destroyed.

Data safety/security:

All information will be stored/obtained under GDPR guidance and treated confidential with participant anonymity adhered to. Good Clinical Practice and local data-protection training will be undertaken.

Success criteria

Success intended	Risk	Contingency plan
Adequate/complete survey responses.	Inadequate/incomplete data.	Option to open survey for further period. Targeting dietitians through my BDA networks, surgeons through Professor Brennan's and speech-therapists through Dr Govender's. Following councils will disseminate free-of-charge: BAHNO, BAHNO-Nurses, BDA. Mandatory responses to prevent closing unless completed.
Meeting recruitment target for cohort study.	Incomplete recruitment.	Ethical approval by 10-months to enable prolonging duration if required.
Successful analysis of cohort data.	Limited experience conducting regression analysis.	Regression training and statistician to advise me on building the regression model.
Meeting recruitment target for interviews.	Not meeting target.	Opening staff-study nationally and offering remote option. Disseminating via professional groups, social-media, charities/support groups. Costed for interpreters to reach all patients.
Running/analysing qualitative workstream.	Limited experience of rapid-methods.	Training courses and Professor Vindrola-Padros's expertise in rapid qualitative research.
Running stakeholder consultation.	Managing different opinions.	Negotiating/consensus training. Dr Govender is experienced in managing stakeholder groups and will advise/support.
Completing on-time.	Multiple workstreams, unanticipated setbacks.	WS1/2 and WS3/4 will run parallel. Rapid approaches used.

Co-producing article.	No experience of coproduction, time required.	Support from UCLH-BRC PPI-team. Designing/supervising as an MSc dissertation if time-limited.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

I attended workshops through UCLH NIHR BRC from April to December 2022 on what PPI is and how to find/do PPI. I was awarded a BRC PPI starter-grant (£300) in open competition (August 2022) for this research to form a PPI group and run an initial group discussion.

I reached out to Barbara Reed, founder of Young Tongues, a support group for people with tongue cancer who represent the majority affected by my research. Barbara disseminated my PPI advert and advised me to meet face-to-face due to impact of surgery on speech intelligibility and understanding over a virtual/hybrid model. I carefully selected five patients from my clinical caseload and Young Tongues, to ensure representation across ethnicity (two Asian, three White-British), occupation, age (26-74years), gender (three women, two men), disease/flap characteristics. Ethnicity was particularly important as my research is about eating/drinking post-surgery, and cultural variation exists in diet.

PPI feedback included:

Feedback	Actions/Changes
Advised research is meaningful to patients, stating they were 'desperate' to eat/drink after surgery but wanted clarity on long-term benefit.	Improved swallowing and nutritional outcomes made more explicit in lay summary.
Advised a 1-week period is acceptable to record dietary intake but using rough estimates than weighed intakes. To consider hospital meal availability as suitable options were not always provided, impacting on meeting nutritional needs.	Hospital meal availability included in cohort study.
Verified it was feasible to record weight but to provide scales.	Scales costed in application.
Consider patient confidence with transitioning to oral feeding as many felt isolated and nervous, preventing swift transition.	Patient confidence incorporated into interview questions.
Verified staff to survey/interview are dietitians, speech therapists, surgeons, and nurses. To consider Intensive Care staff as oral feeding sometimes commenced there.	Intensive care staff added to inclusion criteria.
Advised to capture whose responsibility decision-making for commencing oral feeding is, as conflicting advice was given between staff which could confuse patients.	Questions regarding decision makers incorporated in staff survey/interview questions.
Change lay summary terms to reflect patient experience more accurately.	Changed: 'ability to chew' to 'amount of swelling in the mouth'. Added: 'being able to confidently swallow safely'.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

During recruitment for my PPI group, I discussed that long-term engagement was desired and all have expressed desire to remain involved. They will be involved at every stage of my research which has started at the research design and proposal stage, with plans to continue through preparation, recruitment, data collection, analysis, dissemination and evaluation. In line with NIHR Involve recommendations, £11,712 is requested to cover payment for time for activities, associated subsistence, travel, childcare and equipment costs. The activities are:

Activity	Description of proposed actions
Training	1) PPI members will be asked to complete free online training provided by NIHR Involve (Public Reviewers: Raising Research Quality) to help prepare and enable a clear understanding of their role and develop their knowledge/skills on key research principles. 2) I will also direct to useful resources e.g. NIHR starting-out guide and PPI toolkit.
PPI feedback groups	1) Recap of the research aims, objectives and plan. Feedback on scoping review search terms e.g. grey literature searches.

	2) Advise on my survey e.g. checking the questions to ensure these will capture variation in practices. 3) Advise on outcome measures for cohort study e.g. patient recorded diet diaries. 4) Advise on my qualitative interviews e.g. developing topic guide, checking questions are appropriately worded, targeted and understood. 5) Review qualitative transcripts to capture patient viewpoints and improve validity. 6) Discuss results of all studies and ideas for toolkit design. Advice on strategies to disseminate results to the patient group.
Reviewing documents	1) Ethics application 2) Recruitment adverts 3) Participant information sheets and diet diaries 4) Toolkit prototype 5) Co-produced manuscript draft
Dissemination and evaluation	1) Attendance to *The Swallows* HNC patient conference with my PPI group to share how PPI has influenced my research. 2) Co-produce an article about the impact of PPI involvement in a co-production journal e.g. *BMC Research Involvement and Engagement.* 3) 2 PPI members to attend stakeholder consultation to review toolkit prototype.

For all activities, I will record all feedback and communicate back to the group with suggested amendments. If suggested changes are not made, I will feedback why these were not feasible.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable

8. Training & Development and Research Support

Proposed training and development programme

My existing research experience helped me develop skills in quantitative research, trial delivery, dissemination, and academic writing for publication. The Vitae-RDF helped me identify gaps in my learning to deliver the proposed studies alongside clinical practice training and placements. My training plan supports my long-term aspirations of becoming a Consultant Clinical Academic Dietitian leading a research group in perioperative nutrition at UCL and improving ERAS at UCLH. The GANTT chart included indicates key milestones in my training programme.

Proposed formal study and academic training

As a PhD student at UCL, I will access the comprehensive doctoral skills development programme which includes PhD training modules such as preparing a thesis. I will also attend courses on *Data Management, Zotero, Creating Infographics* and *Designing an Impactful Poster* to equip me with secure data collection, referencing management, and dissemination. Whilst I have undergone several PPI workshops, I found facilitating my group challenging and would benefit from further training through undertaking the *Facilitation of PPI activities* workshop.

To develop skills in scoping review methodology, I will attend the online *JBI Scoping review workshop* at University of Nottingham and access 1:1 support from a research librarian at UCL to develop my search strategy. I will also undertake two modules at UCL in *Survey Design and Quantitative Survey Research* to equip me with design, conduct and analysis of survey data.

I have previously undertaken quantitative research and completed the UCL Practical Medical Statistics Course during my pre-doctoral fellowship which has provided a good baseline training in statistics and software (SPSS and Stata). I will undertake the following courses at UCL to further develop these skills for analysing my cohort study data: *Statistics for observational studies*, *Regression Analysis and Introduction to R*. I have prior experience using Diet-Plan software for analysing diet diaries, but plan to use Nutritics as this is more robust. Professor Nathan Davies (Dietetics programme lead at UCL) has agreed to provide me with a license free-of-charge and attend his practical classes using this software.

Qualitative research methods comprise a key component of this project and I will invest time in developing and expanding my knowledge and skills in this area. To develop baseline knowledge, I will attend the *Qualitative Research Methods in Health* short-course at UCL which includes practical skills for conducting interviews. I will also undertake the *Rapid Research Evaluation and Appraisal Bootcamp* at UCL to equip me with using the rapid methods I am proposing. Having not conducted any previous participant observations, I will attend the *Ethnography for Healthcare Improvement Summer School* at University of Leicester. These courses do not include training on software for coding/analysis, therefore I will also undertake the *Introduction to NVivo* short-course at University of Surrey. Lastly, whilst I have published quantitative research, I have not published qualitative research therefore I will attend the *Writing/Publishing Qualitative Research* course at UCL.

Finally, central to my research is the development of a toolkit anticipated to be a complex intervention drawing on implementation science theories. I will therefore attend the *Mixed-Methods approaches* course at UCL, *Developing and Evaluating Complex Interventions,* online-course at University of Glasgow and the *Implementation Science Masterclass,* at KCL to develop my knowledge of implementation planning, development, and evaluation. I will also undertake the *Influencing and Negotiating* course at UCL to effectively manage my stakeholder consultation group.

Clinical academic development

Extend skills in Enhanced Recovery After Surgery (ERAS)

My research and clinical interests align well with the topic of ERAS. Current ERAS Centres of Excellence are mainly led by colorectal surgeons. A gap exists for an HNC-led Centre of Excellence – it is my aspiration to play a leading role in such a centre at UCLH.

To achieve this, I will attend the *ERAS Society Course* to develop my knowledge of key ERAS principles and visit two leading/founding ERAS Centres of Excellence in Stockholm. This includes the Karolinska Institute and Ersta-Diakoni Hospital, under the supervision of Anna Garpenbeck (ERAS coordinating nurse at Ersta-Diakoni and a board member of the Swedish ERAS Society). I will also visit Professor Bill Fawcett and Dr Chris Jones (Consultant Anaesthetists and ERAS Society committee members) at the Royal Surrey Hospital to learn how to set up an ERAS Centre of Excellence in the UK/NHS.

To develop my wider networks and knowledge of perioperative care, I will attend weekly UCL Centre of Perioperative Medicine multi-disciplinary research/education collaborative meetings. I will shadow flap-surgery and pre-operative anaesthetic assessments to develop my anatomy/physiology knowledge. Finally, I will shadow UCLH Prehabilitation team pre-operative dietetic clinics which are underpinned by ERAS principles.

Developing as a clinical research leader

To support development as both a clinical and research leader, I will enrol on the Rosalind Franklin Programme NHS Leadership Academy online-course. This aligns with my current leadership experience level and is achievable alongside my PhD. I will also attend the *Women in Research Masterclass* at UCL to help me promote gender parity and champion female representation in HNC research, as women are currently underrepresented in this field[74].

To grow the research agenda for dietitians I will attend monthly UCLH Dietetics Research Group meetings and support Dietetic MSc students and staff with research, audit and service evaluation. I also have 1:1 time agreed with Dr Rebecca Carpenter (Clinical Research Manager) at Nutricia Medical, Danone to learn about how industry works alongside the NHS to run clinical nutrition trials.

To learn how a research strategy/culture for nursing and allied health is led, I will work under the mentorship of Dr Rachel Taylor who directs the UCLH Centre for Nursing, Midwife and Allied Health Research. I will support delivery of the internship programme and join the *'Connect'* journal editorial-team.

Ongoing professional practice

I will work one day/fortnight at UCLH alternating outpatient/inpatient work, supervising dietetic students on clinical placement and attending monthly HNC dietetic clinical governance meetings. This will maintain my clinical skills as a specialist dietitian. I will keep up with quarterly BDA HNC subgroup and BAHNO committee meetings and contribute by reviewing documents, developing resources, planning study days and conference organisation.

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Professor Ramani Moonesinghe

Institution	University College London

Position	Head of Research Department for Targeted Intervention, Division of Surgery + Interventional Science

Description of support provided
Professor Ramani Moonesinghe OBE is Professor of Perioperative Medicine at UCL and Honorary Consultant in anaesthetics, perioperative and critical care at UCLH. She has held several prominent leadership positions including: Director, Royal College of Anaesthetists' Health Services Research Centre (2016-22); academic training advisor, National Institute for Academic Anaesthesia (2012-19); training programme director for academic anaesthesia in London (2012-18); Director NIHR Central London Patient Safety Research Collaboration (2023-28); theme lead, NIHR UCLH Biomedical Research Centre (2022-27); NHS England's National Clinical Director for Critical and Perioperative Care (2020-). At UCL, she is group lead, Centre Lead and head of the Research Department for Targeted Intervention where my PhD will be based. She currently has six doctoral students; she has previously supervised six clinicians through to PhD completion; five more are writing up. She has also been a mentor / advisor to NIHR-funded early/mid-career researchers in Bristol, Birmingham and London. She has a longstanding relationship with both Professor Vindrola-Padros and Dr Govender, both of whom she has supported to academic promotion. She leads research which uses multiple methodologies including quantitative, qualitative, clinical trials, mixed-methods evaluation, complex intervention development and implementation

science. Aligned to this proposal, an approach to enhanced recovery for which she developed the main evidence has now been adopted by the NHS and supported by NHSE financial incentives.

Formal supervision meetings with all three supervisors will occur monthly. In keeping with UCL guidance I will also have a thesis committee to ensure that I have consistent external evaluation and support.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Dr Roganie Govender MBE PhD FRCSLT

Institution	University College London and University College London Hospitals NHS Foundation Trust.

Position	Associate Professor (UCL) Consultant Clinical Academic Speech and Language Therapist (UCLH)

Description of support provided
Dr Govender is a Consultant Clinical-Academic SLT, an NIHR Clinical-Lecturer and Associate Professor at UCL. She has over 35 peer-reviewed publications and is a regular invited speaker at national and international academic conferences. She is Chief-investigator on a NIHR PhaseII multicentre dysphagia trial, chair of the NIHR Speech Therapy in ENT Group (sub-group of CRN-ENT National Strategy Group), previous NIHR SLT training advocate, and an RCSLT research advisor. Dr Govender is currently on the supervisory panel for 3 ICA Doctoral Fellows with the first due to complete next year. She has supervised 3 PCAFs to completion (all remaining on the research trajectory). In addition to this, she is supervisor to 4 current awardees on NIHR/HEE infrastructure awards via CRNs or BRCs and one post-doctoral awardee of the NIHR Knowledge and Skills Enhancement award. Dr Govender has been formally supervising my clinical academic development since 2021, during my HEE/NIHR ICA pre-doctoral bridging programme, and UCLH pre-doctoral Chief Nurse Research Fellowship. I have built an excellent clinical and academic relationship with Dr Govender since my employment at UCLH in 2015. Her SLT clinical experience will be vital for the toolkit proposed. Dr Govender's research work straddles the topics of dysphagia, health behaviour change and health service delivery, all of which will contribute to my own project. She has experience of both qualitative and quantitative research methods having spent time and received training within UCL Department of Behavioural Science and Health, and the UCL Department of Applied Health Research. Supervision will occur monthly.

ADDITIONAL ACADEMIC SUPERVISOR 2

Name	Professor Cecilia Vindrola-Padros

Institution	University College London

Position	Professorial Research Fellow, Division of Surgery and Interventional Science

Description of support provided
Professor Vindrola-Padros is a Medical Anthropologist and Professorial Research Fellow in the Department of Targeted Intervention at UCL. She is Director of the Rapid Research Evaluation and Appraisal Lab (RREAL) and

leads a team of 25 researchers interested in applied health research.

Professor Vindrola-Padros's research themes include cancer, behaviour change, global health and migration with interests in applied health research, delivery of cancer treatment and development of rapid approaches to research. She has written extensively on the use of rapid qualitative research and directs the Rapid Research Evaluation and Appraisal Lab at UCL. She has supervised 10 PhD students to date with 3 PhD completions.

Professor Vindrola-Padros has expertise in ethnographic research in healthcare, rapid qualitative research and participatory and collaborative research. She is the past director of the Qualitative Health Research Network (QHRN) and a fellow of the Society for Applied Anthropology. Professor Vindrola-Padros's expertise in qualitative research, rapid approaches and developing new models of care aligns with this research proposal which includes a qualitative workstream using interviews and observations, and development of an intervention in cancer.

Supervision will occur monthly, joint with Professor Moonesinghe and Dr Govender either face-to-face or via MS teams and email communication in between. More frequent 1:1 supervision (weekly to fortnightly) will be provided during the qualitative workstream.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Alastair Duncan

Institution	Guys and St Thomas' NHS Foundation Trust; King's College London; King's Academic Health Partners

Position	Consultant Clinical Academic Dietitian and Lecturer in Nutrition and Dietetics

Description of support provided
Dr Duncan is a Consultant Clinical Academic Dietitian and Lecturer in Nutrition and Dietetics employed by both Guys and Thomas' NHS Foundation Trust and King's College London . He completed an NIHR Clinical Doctoral Research Fellowship in 2016 in Nutrition and Infectious Diseases with clinical research interests in health inequalities, ageing, HIV care, obesity, diabetes, and qualitative research methodologies. Dr Duncan has provided me with excellent support during my pre-doctoral fellowship and with developing this application. He has agreed to provide formal mentorship on a bimonthly basis with informal meetings between as required. Dr Duncan has been invaluable in recommending training courses, broadening my thinking of different methodological approaches, and introducing me to research dietitians in his network. Dr Duncan has successfully developed a role as a Consultant Clinical Academic Dietitian delivering on each of the four pillars of advanced clinical practice (clinical practice, leadership, research and education). He has secured a £10 million award from the Southeast London Integrated Care Board to fund clinical care for people living with complex obesity and research addressing health inequalities these patients experience through his team of PhD and MSc researchers. He is well-placed to support me with navigating my NIHR journey and accessing opportunities towards my future career aspirations of becoming a Consultant Clinical Academic Dietitian. Dr Duncan has a track record of successful mentorship and is able to support me through the challenges that I may encounter during my DCAF and development as a Consultant Clinical Academic Dietitian.

PRACTICE SUPERVISOR 2

Name	Professor Peter Brennan

Institution	Portsmouth University Hospitals NHS Trust

Position	Consultant Oral and Maxillofacial Surgeon and Honorary Professor of Surgery

Description of support provided
Professor Brennan is a Consultant Oral and Maxillofacial Surgeon with clinical interests in HNC reconstructive free-flap surgery and patient safety. He has an MD in cancer biology and a PhD in human factors. Professor Brennan was awarded a personal chair in surgery in recognition of his research and education profile, has supervised 6 PhDs and examined many MDs and PhDs. He has contributed to several surgical textbooks, lead editor of Gray's Surgical Anatomy and editor-in-chief of *Journal of Oral Pathology and Medicine*. Professor Brennan is integrated internationally with positions that include: Elected Council Member of the Royal College of Surgeons, British Association of Oral and Maxillofacial Surgeons and BAHNO council. My research proposal includes development of a post-operative feeding toolkit in patients with HNC. Surgeons will be key users of this toolkit and it is vital that I have a surgeon involved in my research, for example, by advising me on outcome measures for my cohort study pertaining to disease and surgical characteristics that influence eating/drinking after surgery. Professor Brennan is well placed to support me with my research given his clinical and research background and interests. I have developed an excellent working relationship with him through working together on BAHNO council for three years and co-authoring publications including national standards. He will provide ongoing support as a surgical advisor on my research, dissemination and mentor by helping me develop my professional networks in HNC which is crucial for conducting future multi-centre studies and implementing my research within surgical units.

Collaborations
Professor Olle Ljungqvist Professor of Surgery at Orebro University Hospital, Sweden and an international expert in ERAS. He chaired the European Society for Clinical Nutrition and the International Association for Surgical Metabolism and Nutrition. Professor Ljungqvist co-founded the ERAS Society, has successfully led and implemented novel ERAS research internationally and contributed to multiple ERAS consensus guidelines. Professor Ljungqvist has been advising me and will provide ongoing support on the design of my research from an ERAS perspective to optimise future implementation alongside supporting my professional development by organising an overseas placement in Stockholm visiting two ERAS centres of excellence to develop the skills, knowledge and networks to achieve my aspirations for ERAS at UCLH. **Clinical Associate Professor Merran Findlay** Executive Research Lead (Cancer Nutrition) at Prince Alfred Hospital and Chris O'Brien Lifehouse Cancer Centre in Sydney, Australia. An Advanced Accredited Practising Dietitian, she holds two current fellowships as Research Translation Fellow, University of Sydney and Senior Research Fellow, University of New South Wales. A/Prof Findlay has expertise in nutritional assessment methods and has advised me on most appropriate outcome measures for my cohort study. Furthermore, A/Prof Findlay has expertise in implementation science and integrating evidenced-based nutrition care guidelines into practice, advising me on my research from an implementation perspective and will provide ongoing support. A/Prof Findlay has led international HNC dietetic guidelines, contributed to HNC ERAS consensus guidelines and implementation science networks in Australia and will support me with building networks and implementing my findings. **Professor Gareth Ambler** Professor in Medical Statistics at UCL with research interests and expertise in regression analysis and developing prediction models. Professor Ambler has been advising me on the statistical analysis plan for my cohort study and has been costed into my application to provide ongoing support during my PhD.

Ms Ursula Mackie-Savage

HNC ERAS Nurse at UCLH and leads on implementing and auditing ERAS protocols. Ursula will support me with implementing my findings locally.

British Association of Head and Neck Oncologists (BAHNO)

National HNC organisation and a gatekeeper for endorsing recommendations. BAHNO and council members in major UK HNC units will assist with advertising my survey to their membership which includes key decision-makers for post-operative feeding practices.

British Dietetic Association (BDA) HNC group

I sit on the committee which has dietetic representatives from major UK HNC centres nationally and will act as key links for gaining survey responses. This includes Mr Long Li, dietetic representative on NHS England Getting It Right First Time (GIRFT) HNC workstream which aims to review UK HNC service delivery and identify/reduce variation in care. Long will review my survey questions and support me with accessing HNC unit leads alongside implementing my findings.

Nutricia Medical, Danone

International market leader in clinical nutrition products including specialist ERAS formulas utilised at UCLH and major funder of clinical nutrition trials. I have 1:1 time agreed with Dr Rebecca Carpenter (UK Senior Clinical Research Manager) to learn about industry partnerships where future research collaboration and funding could be sought.

UCLH Centre for Nursing, Midwife and Allied Health research

A community who has supported me on my clinical academic journey through mock interviews, my pre-doctoral fellowship, monthly learning sets and peer support.

Young Tongues

A global peer-to-peer support group for people with tongue cancer which is the most prevalent cancer sub-site in my research. Young Tongues have supported me with advertising for PPI members and will provide ongoing support by advertising and disseminating my research to patient networks.

UCL HNC Academic Centre

Academic arm of UCLH HNC Centre with monthly seminars that reach wide international audiences. Will support me with networking and disseminating my research.

Host Organisation Support Statement
The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support
NHS Host organisation University College London Hospitals NHS Foundation Trust (UCLH) is a member of the Shelford Group which is a collaboration of the ten largest teaching and research NHS hospital trusts in England. As a Trust featuring in the top 10 for research activity, we are committed to embedding research capability and capacity into all health professions, irrespective of grade. This is reflected in the UCLH Strategy for 2023-27, in which we commit to fostering clinical academic careers within nursing, midwifery and allied health professions (NMAHP). We aim to achieve the national target of 1% of NMAHP in clinical academic roles by 2030 through identifying NMAHPs who have the potential to thrive in this role and by supporting them with their career progression. Florence is one of our

staff we are investing in, to become a leader within head and neck dietetics. Florence is currently completing a Chief Nurse Fellowship programme, similar to the NIHR PCAF, in time and on budget according to her planned programme of work.

UCLH is one of the founding members of the biomedical research centres and continues to lead on ground-breaking advances in research for the benefits of patients and sharing best practise with its academic partner UCL. AHP clinical academics are essential for BRC funded institutions. Florence's proposed fellowship will be based across UCL and UCLH, aligning with UCLH Strategy for 2023-27 to deepen collaboration between UCLH/UCL and broaden the workforce and career opportunities for NMAHPs to improve the diversity in expertise and knowledge across different healthcare professionals. This aligns with the recent NHS Long-term workforce plan which includes developing NMAHP clinical academics to bridge healthcare providers with academic institutions and improve the diversity and inclusion of our underrepresented groups in all aspects of healthcare including research.

Should Florence's application be successful, we will continue this support through our peer support programme for doctoral and clinical academic NMAHPs. She will have access to the CNMAR Research Education Framework, incorporating workshops and lectures delivered by clinical academic staff to equip NMAHPs with the practical skills and ethical guidance for conducting research with integrity in complex, healthcare settings. Florence will also be encouraged and supported to access this additional training, which aims to apply theoretical research training into clinical practice.

Members of our peer support programme are ambassadors for inclusion and equality in research and our commitment to diversity in research is discussed in all meetings to inform new collaborations and research outputs. UCLH is certified by the International Disabilities Management Standards Council. Delivery of equality objectives are monitored and supported by the Diversity and Equality Steering Group. Members include representatives from the Black Asian Minority Ethnic (BAME) network. The BAME and LGBTQ+ networks focus on empowering staff to achieve their potential, influencing UCLH policy and supporting progression in the workforce. We adhere to the NHS Equality Delivery System 2 (EDS2) to support inclusive leadership and promote workforce race equality standards (WRES).

UCLH will comply with the relevant Principles and Obligations for clinical academic training.

HEI partner organisation

Florence has demonstrated her commitment and dedication towards a clinical academic career through the development of this proposal which is rooted in her clinical dietetic background in head and neck cancer. She has exceptional potential, evidenced by her strong academic achievement at both undergraduate and postgraduate level and outputs during her clinical career and winning two pre-doctoral fellowships.

Florence will be based in the Research Department for Targeted Intervention (RDTI), within the Division of Surgery and Interventional Science (DSIS), in the Faculty of Medical Sciences at UCL. Our department provides an excellent environment for early and mid-career researchers, particularly clinical academics. UCL, and specifically DSIS, is committed to promoting equality, diversity and inclusion, and holds a silver Athena Swan award. Within DSIS, we currently have over 100 doctoral research students, and RDTI hosts the largest proportion of these across our three campuses. RDTI is host to seven thriving research centres, each including clinician and career scientists, covering a range of research portfolios from discovery science through to policy and implementation research. Our team includes several clinician scientists from allied health professions who will be able to provide mentorship to Florence as she progresses her studies. RDTI currently hosts several awards from NIHR and UKRI including an NIHR Research Professor, an MRC Senior Fellow and, in conjunction with UCLH, the NIHR Central London Patient Safety Research Collaboration (CL-PSRC). Furthermore, the RDTI leadership team include two theme leads at the UCLH NIHR Biomedical Research Centre, demonstrating our strong working relationship with our main NHS partner.

Florence's proposal is well aligned with the strategic direction of the division. She will specifically be based within the UCL Head and Neck Academic Centre (HNAC) and the Centre for Perioperative Medicine (CPOM) both based at RDTI. HNAC is the academic arm of the UCLH head and neck cancer unit, aligning with Florence's clinical speciality. Florence will also be supported by the NIHR CL-PSRC, with access to the PSRC learning academy which currently includes five doctoral and three post-doctoral scientists, and which provides mentoring and peer support from a multidisciplinary team including frontline clinicians, policy makers and academics with a diverse skill set. The Rapid Research Evaluation and Appraisal Lab (RREAL) also sits within RDTI, and is led by Professor Vindrola-Padros, Florence's secondary supervisor. RREAL is uniquely placed to support Florence with the rapid methodologies she is proposing to undertake. UCL also has a rapidly growing Nutritional Sciences department within the Division of Medicine which recently opened a pre-registration MSc degree in Dietetics (2021). Florence will be able to access the expertise, training and equipment available from this department to support her during her DCAF. Finally, Florence can exploit the extensive skills development programme for staff and students at all levels at UCL, through the UCL doctoral school, and the wider offer from UCL with over 170 live or self-paced courses encompassing: leadership; engagement; influence and impact; knowledge and intellectual abilities; personal effectiveness; and research governance.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Florence Cook

Employing organisation	University College London Hospitals NHS Foundation Trust

Grade	Band 7

Starting point on scale	50056

Increment Date	

Justification of Costs
Salary costs for 1.0 WTE Band 7 Dietitian over the 3 years of this award

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	50056
Geographical Weighting (£)	7746
Other allowances (£)	
Superannuation (£)	8312
Nat. Insurance (£)	6721
Current Annual Costs (£)	**72835**

2

Staff type	Shared Staff

Name	Professor Gareth Ambler

Employing organisation	University College London

Role on this research	Statistician

Grade	Grade 10/1

Starting point on scale	Point 1

Increment Date	01/08/2024

Justification of Costs
Professor Ambler is Professor in Medical Statistics at UCL with research interests and expertise in regression analysis and developing prediction models. Professor Ambler has been advising me on the statistical analysis plan for my cohort study in this proposal. This includes complex regression analysis which goes beyond the scope of a typical PhD student. I will require his expertise and advice during the the quantiative workstream in my PhD. 8% for 6 months costing is equivalent to 2 weeks of his time, which will be spread out during the analysis phase of my cohort study.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	75308
Geographical Weighting (£)	4500
Other allowances (£)	15000
Superannuation (£)	20478
Nat. Insurance (£)	11827
Current Annual Costs (£)	**127113**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Miss Florence Cook
Type of cost: NHS
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Lead applicant salary costs for 3 years

	Year 1	**Year 2**	**Year 3**	**Total Cost (£)**
Basic Salary	54662	57122	59693	171477
National Insurance	8185	8996	9878	27059
Superannuation	11304	11813	12344	35461
Geographical weighting / Other allowances (if applicable)	7746	7746	7746	23238

Staff Type: Other
Other Type: Shared Staff
Name: **Professor Gareth Ambler**
Type of cost: HEI
% Full-time on this research: 8.0
Total months on this research: 6.0

Justification of Cost: Professor Ambler will support me with statistical analysis of my cohort study during the third year of my PhD. 8% for 6 months costing is equivalent to 2 weeks of his time which will be spread out during the 6 months.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	0	0	5581	5581

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	0	0	4465	4465
Total NHS	257235	0	0	257235
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: British Association of Head and Neck Oncologists Annual Scientific meeting
Justification of Cost: British Association of Head and Neck Oncologists conference fee (allied health/affiliate fee).

	Year 1	Year 2	Year 3	Total Cost (£)
	75	0	0	75

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: British Association of Oral and Maxillofacial Surgeons conference fees
Justification of Cost: British Association of Oral and Maxillofacial Surgeons conference fee (early-bird fee, daily attendance= £250).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	250	0	250

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: Enhanced Recovery After Surgery World Congress
Justification of Cost: Enhanced Recovery After Surgery World Congress including ERAS course fees. $395 /£344 student early bird fee.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	344	0	344

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: British Dietetic Association Research Symposium fees
Justification of Cost: British Dietetic Association Research Symposium fees (£60 member fee).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	60	60

Type of cost: NHS
Description: Conference Fees
Conference: True

Item Description: The Swallows head and neck cancer patient conference
Justification of Cost: Free admission.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: ERAS World Congress accommodation
Justification of Cost: ERAS World Congress location yet to be determined but based on past locations (Madrid, Lisbon, Liverpool, Stockholm, Glasgow, Valencia, New Orleans) estimates from HMRC gov.uk (https://www.gov.uk/guidance/expenses-rates-for-employees-travelling-outside-the-uk) range from £75 to £160/night. Taken mid-point of £120/night which is in keeping with several European cities. 3 days, 4 nights hotel = (£120/night)= £480.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	480	0	480

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: ERAS World Congress subsistence
Justification of Cost: Lunch provided at 3 day conference but nil else. Subsistence costs estimated from gov.uk: (https://www.gov.uk/guidance/expenses-rates-for-employees-travelling-outside-the-uk) guidance based on the range of prior countries (most to least expensive). 3 days subsistence at £35/day (£105 total) estimated as average of possible countries at £25/day (UK-least expensive) or 917.3 Swedish Krona equivalent to £66/day (Sweden-most expensive).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	105	0	105

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: ERAS World Congress travel costs
Justification of Cost: Flights for ERAS World Congress - location yet to be determined but based on prior locations (maximum = USA) £920 return. Return train ticket to airport in London (£24, National Rail fare) and in conference city (£40).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	984	0	984

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: The Swallows patient conference travel costs
Justification of Cost: Rail fare based on average of previous locations (Sunderland and Torquay) taken from National Rail quoted fares. Return rail fare from London = £114.80.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	115	115

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: British Dietetic Association research symposium travel costs
Justification of Cost: For travel to and from British Dietetic Association research symposium. Return journey taken National Rail quotes fares: London to Birmingham= £96.20.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	96	96

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: The Swallows patient head and neck cancer conference accomodation.
Justification of Cost: Accommodation (based on previous locations, Sunderland and Torquay) for 2 nights estimated at £75/night = £150. Estimate taken from HMRC gov.uk guidance (accommodation for UK location outside London: https://www.gov.uk/government/publications/marine-licensing-fees/travel-and-subsistence-costs).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	150	150

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: The Swallows patient head and neck cancer conference subsistence
Justification of Cost: The Swallows patient head and neck cancer conference. 2 days subsistence at £25/day = £50 estimated from gov.uk (https://www.gov.uk/government/publications/marine-licensing-fees/travel-and-subsistence-costs#subsistence).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	50	50

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Clinical Placement at Royal Surrey Hospital
Justification of Cost: 3 days subsistence for clinical placement at £25/day per gov.uk guidance (https://www.gov.uk/government/publications/marine-licensing-fees/travel-and-subsistence-costs#subsistence) = £75.

	Year 1	Year 2	Year 3	Total Cost (£)
	75	0	0	75

Type of cost: NHS
Description: Journey Costs
Conference: False
Item Description: Travel costs for Clinical Placement at Royal Surrey Hospital
Justification of Cost: Return rail fare from London to Guilford (£32.20) x 3 days =£96.60 (quote from National Rail). Taxi from Guilford railway station to Royal Surrey Hospital (£8) x 6 journeys = £48.

	Year 1	Year 2	Year 3	Total Cost (£)
	145	0	0	145

Type of cost: NHS
Description: Journey Costs
Conference: False
Item Description: Travel costs for University of Leicester Ethnography for Healthcare improvement course
Justification of Cost: Return rail fare from London to Leicester = £74.50 (quote from National Rail).

	Year 1	Year 2	Year 3	Total Cost (£)
	75	0	0	75

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Accommodation for University of Leicester Ethnography for Healthcare improvement course
Justification of Cost: 2 nights accommodation in University of Leicester College Court= £84/night x 2 nights = £168.

	Year 1	Year 2	Year 3	Total Cost (£)
	168	0	0	168

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Subsistence for University of Leicester Ethnography for Healthcare improvement course
Justification of Cost: Course fees include lunch but not other meals. 2 days of 2 meals subsistence at £10/day = £20 (estimated from gov.uk: https://www.gov.uk/government/publications/marine-licensing-fees/travel-and-subsistence-costs#subsistence).

	Year 1	Year 2	Year 3	Total Cost (£)
	20	0	0	20

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Accommodation for Overseas Research Visit in Sweden
Justification of Cost: 5 nights accommodation at 1939SEK (Swedish Krona) equivalent to £140/night in Stockholm (£700). Estimates taken from gov.uk: https://www.gov.uk/guidance/expenses-rates-for-employees-travelling-outside-the-uk#countries-s-to-u.

	Year 1	Year 2	Year 3	Total Cost (£)
	700	0	0	700

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Subsistence for Overseas Research Visit in Sweden
Justification of Cost: 5 days subsistence at 917.30SEK (Swedish Krona) equivalent to £66/day (£330). Estimates taken from gov.uk: https://www.gov.uk/guidance/expenses-rates-for-employees-travelling-outside-the-uk#countries-s-to-u.

	Year 1	Year 2	Year 3	Total Cost (£)
	330	0	0	330

Type of cost: NHS
Description: Journey Costs
Conference: False
Item Description: Travel for Overseas Research Visit in Sweden
Justification of Cost: Return flight to Stockholm (£229). Estimate taken from Skyscanner. Return rail fare to airport in London (£24). Estimate taken from National Rail. Return rail fare from Stockholm airport to city centre (£35). Day travelcard to travel between hotel to hospital sites and between. 7-day travel card is cheaper than costing 5 days and costs 420SEK (Swedish Krona) equivalent to £31. Estimate taken from https://sl.se/en/in-english/fares--tickets/visitor-tickets/travelcards/.

	Year 1	Year 2	Year 3	Total Cost (£)
	319	0	0	319

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Overseas Research Visit in Sweden
Justification of Cost: To use phone whilst outside UK to access emails and co-ordinate daily activities with surgical units/staff. My provider EE quote £2.29/day to use minutes/texts/data allowance in Sweden. £2.29x5 = £11.45.

	Year 1	Year 2	Year 3	Total Cost (£)
	11	0	0	11

Type of cost: NHS
Description: Journey Costs
Conference: False
Item Description: Travel costs to attend University of Surrey NVivo course

Justification of Cost: Return rail fare from London to Guilford (£32.20) x 2 days =£64.40 (quote from National Rail).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	64	0	64

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	1918	2227	471	4616

Equipment

Type of cost: NHS
VAT relief to be applied: True
Description: Encrypted Dictaphone (Olympus DS 9000 system edition)
Price excluding VAT £: 245
Expected life time: 36
Justification of Cost: To record qualitative interviews and PPI consultation group discussions .

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	294	0	0	294

Type of cost: NHS
VAT relief to be applied: True
Description: Encrypted USB memory stick (Integral, 64GB Secure 360)
Price excluding VAT £: 20
Expected life time: 36
Justification of Cost: To back up data collection securely.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	24	0	24	48

Type of cost: NHS
VAT relief to be applied: True
Description: Laptop
Price excluding VAT £: 1000
Expected life time: 60
Justification of Cost: To use during my PhD and whilst travelling from placement sites. I plan to purchase an Apple MacBook Air (M2 chip) as this is a laptop I currently use for academic work and is robust in terms of longevity and for avoiding viruses. This model costs £1248 (inc VAT) / £1040 (exc VAT). As DCAF applicant notes specify that up to £1000 (exc VAT) / £1200 (inc VAT) can be expensed for a laptop, I will make up the remainder of the cost personally as it is my preference to use this particular laptop.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	1200	0	0	1200

Type of cost: NHS
VAT relief to be applied: True
Description: Tablet and smart pencil
Price excluding VAT £: 590
Expected life time: 60
Justification of Cost: For recording legible fieldnotes in observations and taking notes during interviews and PPI groups. Apple IPad mini tablet 6th generation (£569 inc VAT) and Apple Pencil 2nd generation (£139.00 inc VAT).

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	708	0	0	708

Type of cost: NHS
VAT relief to be applied: True
Description: Electronic bathroom scales
Price excluding VAT £: 13
Expected life time: 24
Justification of Cost: Salter 9028 SV3R09 Razor Electronic Bathroom Scale, Easy Read (£14.99/unit (inc VAT) / £13 (exc VAT). For cohort study participants (anticipated n=90) to take weight measurements at home. 90 x £14.99 = £1349.10.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	0	1349	0	1349

Type of cost: NHS
VAT relief to be applied: True
Description: Spare batteries for scales
Price excluding VAT £: 6
Expected life time: 12
Justification of Cost: LiCB 20 PCS CR2032 Lithium Coin Battery. £6.99 for 20 batteries. Each scale comes with a battery, but it would be prudent to have spare batteries in case.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	0	7	0	7

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	2226	1356	24	3606

Consumables

Description: Notepads
Type of cost: NHS
Please describe fully: Notepads
Justification of Cost: 8x A4 notepads (2.99 each). Quote taken from Ryman: (https://www.ryman.co.uk/services/copy). 2 packs per year for research team taking notes during all workstreams, PPI consultation groups and stakeholder consultation.

	Year 1	Year 2	Year 3	Total Cost (£)
	6	6	6	18

Description: Pens
Type of cost: NHS
Please describe fully: Pens
Justification of Cost: Pack of 10 bic ballpoint pens (£3.00). Quote taken from Ryman: (https://www.ryman.co.uk/services/copy). 1 pack per year for taking notes during all workstreams, PPI consultation groups and stakeholder consultation.

	Year 1	Year 2	Year 3	Total Cost (£)
	3	3	3	9

Description: Photocopying
Type of cost: NHS
Please describe fully: Photocopying
Justification of Cost: A4 black and white cost for 1000+ pages = 15p/page single-sided (double sided is equivalent to twice the cost). Quote taken from Ryman: (https://www.ryman.co.uk/services/copy). For diet diaries (8 A4 pages/90), cohort study participant information sheets (2 A4 pages/90), interview and observation participation sheets (2 A4 pages/31), PG-SGA forms (2 A4 page/90), Consent forms (1 A4 page/108: 90 for cohort study + 18 for staff interviews). Total: 720+180+62+180+108 =1250 pages x 0.15 = £187.50.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	188	0	188

Description: Large A4 envelopes
Type of cost: NHS
Please describe fully: Large A4 envelopes
Justification of Cost: 1x 100 pack peel and seal A4 envelopes = £15.99. Quote taken from Ryman: (https://www.ryman.co.uk/services/copy). For patients to send back diet-diaries and PG-SGA forms.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	16	0	16

Description: Postage
Type of cost: NHS
Please describe fully: Postage
Justification of Cost: Second class stamp for large letter (251-500g) = £2.40 (quote from Royal Mail). For 100x pre-paid envelopes for patients to send back diet diaries and PG-SGA forms. Provide 10% extra in case further envelopes need sending.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	240	0	240

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	9	453	9	471

Patient and Public Involvement

Description: Subsistence (vouchers for time taken for PPI consultation groups)
Type of cost: NHS
Please describe fully: Subsistence (vouchers for time taken for PPI consultation groups)
Justification of Cost: Each consultation group will be for 2 hrs. There will be 2 x 2hr PPI consultation groups per year for 5 PPI members. Costs to cover time estimated at £25/hr per NIHR INVOLVE guidance: Year 1: 2 x PPI consultation group discussions (2hrs each at £25/hr for 5 PPI members = £500) 1) Recap of research project plan, review scoping review search terms and advise on grey literature 2) Reviewing survey study and questions. Year 2: 2 x PPI consultation group discussions (2hrs each at £25/hr for 5 PPI members = £500) 1) Advise on cohort study outcome measures e.g. diet diary format and instructions/example page 2) Reviewing qualitative study including feedback on interview questions and participant information sheets. Year 3: 2x PPI consultation group discussions (2hrs each at £25/hr for 5 PPI members = £500) 1) Feedback on qualitative enquiry transcripts 2) Discuss results of all studies, design toolkit and discuss dissemination of research findings via a co-produced article and to facilitate presentation skills training.

	Year 1	Year 2	Year 3	Total Cost (£)
	500	500	500	1500

Description: Subsistence - journey costs to attend consultation group discussions and stakeholder consultation
Type of cost: NHS
Please describe fully: Subsistence - journey costs to attend consultation group discussions and stakeholder consultation
Justification of Cost: Transport costs (London zone 1-6 day travel card quoted from TfL= £15.20) to UCLH Macmillan Centre for 6 consultation groups for 5 members (£152 Year 1, 2 and 3). Transport costs (London zone 1-6 day travel card at £15.20) to UCLH Macmillan Centre for stakeholder consultation for 2 members (£30.40 Year 3).

	Year 1	Year 2	Year 3	Total Cost (£)
	152	152	182	486

Description: Refreshments for PPI consultation group discussions and stakeholder consultation
Type of cost: NHS
Please describe fully: Refreshments for PPI consultation group discussions and stakeholder consultation
Justification of Cost: For each PPI consultation group (<5hrs away from home) HMRC advise £5 for one meal per person. NIHR Involve recommend £10-20/head for coffee and buffet lunch. In central London, a light lunch costs £10 minimum/person. Year 1: Lunch for 5 people at £10/head for 2 PPI consultation groups = £100. Year 2:

Lunch for 5 people at £10/head for 2 PPI consultation groups = £100. Year 3: Lunch for 5 people at £10/head for 2 PPI consultation groups = £100. Lunch for 2 people to attend stakeholder consultation = £20.

	Year 1	Year 2	Year 3	Total Cost (£)
	100	100	120	320

Description: Childcare to enable attendance to PPI training, consultation groups and stakeholder consultation
Type of cost: NHS
Please describe fully: Childcare to enable attendance to PPI training, consultation groups and stakeholder consultation
Justification of Cost: None of my PPI members currently require this but as peoples circumstances may change, and if members leave and new people join during my PhD I am estimating childcare costs for approximately 50% of members (2 PPI members) requiring to attend the following: Costs are based on £15/hr NIHR Involve guidance Year 1: 4hrs to attend training (2hrs/2 PPI members). 16hrs for 2 of the consultation groups to accommodate travel time (up to 1hr each way) (4hrs (time) x2 PPI members x 2 consultation groups). Total = 20hrs at £15/hr = £300. Year 2: 16hrs for 2 the consultation groups to accommodate travel time (up to 1hr each way) (4hrs (time) x2 PPI members x 2 consultation groups). Total = 16hrs at £15/hr = £225. Year 3: 16hrs for 2 of the consultation groups to accommodate travel time (up to 1hr each way) (4hrs (time) x2 PPI members x 2 consultation groups). 4hrs for 1x2hr stakeholder consultation to accommodate travel time (up to 1hr each way) for 1 PPI member. Total = 20hrs at £15/hr = £300.

	Year 1	Year 2	Year 3	Total Cost (£)
	300	225	300	825

Description: Carer costs to enable attendance to PPI training, consultation groups and stakeholder consultation
Type of cost: NHS
Please describe fully: Carer costs to enable attendance to PPI training, consultation groups and stakeholder consultation
Justification of Cost: Carer costs for participants based on £20/hr as mid-point for hourly rate and £200/day in NIHR Involve guidance of £7.50-£40/hr and £200/day. Currently none of my PPI group require this but estimating costs if circumstances change at maximum 50% of PPI members requiring this to attend the following activities: Year 1: 8hrs for 2 PPI members to attend training (4hrs/PPI member). 16hrs for each of the 2 consultation groups to accommodate travel time (up to 1hr each way) (4hrs x 2 groups x 2PPI members). Total = 24hrs at £20/hr = £480. Year 2: 16hrs for each of the 2 consultation groups to accommodate travel time (up to 1hr each way) (4hrs x 2 groups x 2PPI members). Total = 16hrs at £20/hr = £320. Year 3: 16hrs for each of the 2 consultation groups to accommodate travel time (up to 1hr each way) (4hrs x 2 groups x 2PPI members). 8hrs for 1x stakeholder consultation (4hrs x 2 PPI members). Total = 24hrs at £20/hr = £480.

	Year 1	Year 2	Year 3	Total Cost (£)
	480	320	480	1280

Description: Subsistence - support worker/personal assistant
Type of cost: NHS
Please describe fully: Subsistence - support worker/personal assistant
Justification of Cost: None of my PPI members currently require this but provision made for 1 PPI member if this is required. Costs based on £9.50/hr per NIHR Involve (OR £200/day). Year 1: 8hrs for 2 consultation groups (2hrs) + 2hrs to accommodate travel time. Total = 8hrs at £9.50/hr = £76. Year 2: 8hrs for 2 consultation groups (2hrs) + 2hrs to accommodate travel time. Total = 8hrs at £9.50/hr = £76. Year 3: 8hrs for 2 consultation groups (2hrs) + 2hrs to accommodate travel time. 4hrs for 1 stakeholder consultation. 2 days for attending conference. 10hrs at £9.50/hr (£95) + 2 days at £200/day (£400). Total=£495.

	Year 1	Year 2	Year 3	Total Cost (£)
	76	76	495	647

Description: Venue to run PPI consultation groups
Type of cost: NHS
Please describe fully: Venue to run PPI consultation groups
Justification of Cost: I have been using the UCLH Macmillan Centre room and I have been advised by the manager of the centre that I can continue to use this room free of charge for my PPI consultation groups, qualitative interviews and stakeholder consultation. UCLH MCC is wheelchair accessible, on the ground floor and served within a 5 minute walk of many national rail and London underground stations.

	Year 1	Year 2	Year 3	Total Cost (£)

	0	0	0	0

Description: Equipment - Office supplies
Type of cost: NHS
Please describe fully: Equipment - Office supplies
Justification of Cost: Based on NIHR Involve guidance at £5 per activity to cover home office supplies. Year 1: To undertake 3 modules at home for PPI training. 5 people x 3 modules = £75.

	Year 1	Year 2	Year 3	Total Cost (£)
	75	0	0	75

Description: International Head and Neck Cancer Conference (The Swallows) fees
Type of cost: NHS
Please describe fully: International Head and Neck Cancer Conference (The Swallows) fees
Justification of Cost: For 5 PPI members to attend patient conference. This conference is free of charge therefore no costs incurred.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Subsistence - accommodation to attend International Head and Neck Cancer Conference (The Swallows) (2 days, 2 nights)
Type of cost: NHS
Please describe fully: Subsistence - accommodation to attend International Head and Neck Cancer Conference (The Swallows) (2 days, 2 nights)
Justification of Cost: Year 3: Accommodation (based on previous locations, Sunderland and Torquay) 2 nights estimated at £75/night for 5 people = £750. Estimates taken from HMRC gov.uk guidance (accommodation for UK location outside London: https://www.gov.uk/government/publications/marine-licensing-fees/travel-and-subsistence-costs).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	750	750

Description: Other costs e.g. Interpretation services, communication support (e.g. hearing/induction loops), mobility aids e.g. walking aids
Type of cost: NHS
Please describe fully: Other costs e.g. Interpretation services, communication support (e.g. hearing/induction loops), mobility aids e.g. walking aids
Justification of Cost: Interpretation services: Quotes from UCLH language line service: Face-Face interpretation = £23/hr. A local audit found 14% of patients with head and neck cancer require interpretation services. None of my PPI members currently require interpretation but I have costed for 1 member in the event of new PPI members who require this. Year 1: 2hrs training + 4hrs for 2 consultation groups + 3hrs to translate 3 documents (1hr/document) for review over telephone language line. Total = 9hrs at £23/hr = £207. Year 2: 4hrs for 2 consultation groups. Total = 4hrs at £23/hr = £92. Year 3: 4hrs for 2 consultation groups + 2hrs for stakeholder consultation per person + 1hr to translate 2 documents for review over telephone language line. Total = 7hrs at £23/hr = £161.

	Year 1	Year 2	Year 3	Total Cost (£)
	207	92	161	460

Description: Recruitment
Type of cost: NHS
Please describe fully: Recruitment
Justification of Cost: I will not require advertising costs. I recruited my PPI group via my personal caseload and support groups for which I gained a lot of interest. I do not anticipate I will require costs to support me with recruiting PPI members.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Involvement training
Type of cost: NHS

Please describe fully: Involvement training
Justification of Cost: I will not require funds for this. UCLH Biomedical Research Centre offers several PPI workshops at regular intervals free of charge for UCLH/UCL staff.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Administrator support
Type of cost: NHS
Please describe fully: Administrator support
Justification of Cost: For processing payment/rewards/expenses: No cost will be incurred as I will manage this during my fellowship and the finance administrator at my host organisation has agreed to support me as part of their role in their division.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Independent facilitator
Type of cost: NHS
Please describe fully: Independent facilitator
Justification of Cost: My supervisory team includes members of staff that are experienced in working with people with head and neck cancer and facilitation of PPI group discussions. I will not require any specific skills to be costed for by use of an independent facilitator.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Disclosure and Barring Service
Type of cost: NHS
Please describe fully: Disclosure and Barring Service
Justification of Cost: Not required as not having direct contact with young people and children or taking part in any interviews.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Subsistence (vouchers for time taken to attend PPI training)
Type of cost: NHS
Please describe fully: Subsistence (vouchers for time taken to attend PPI training)
Justification of Cost: Year 1: 5 PPI attendees to attend training modules at 1-2hrs each (x3 modules) 'Public reviewing' NIHR Involve training-online for the public. Courses are free of charge. Costs to cover time for attending PPI training at £25/hr per INVOLVE guidance for 6hrs total/person for 5 PPI members = £750.

	Year 1	Year 2	Year 3	Total Cost (£)
	750	0	0	750

Description: Subsistence (vouchers for time taken to review documents)
Type of cost: NHS
Please describe fully: Subsistence (vouchers for time taken to review documents)
Justification of Cost: Year 1: 3 documents at £25/1hr task per INVOLVE guidance for 5 PPI members (£375). 1) ethics application 2) recruitment adverts 3) participant information sheets. Year 3: 2 documents at £25/1hr task per INVOLVE guidance for 5 PPI members (£250). 1) Toolkit prototype 2) Co-produced manuscript draft.

	Year 1	Year 2	Year 3	Total Cost (£)
	375	0	250	625

Description: Subsistence (vouchers for time taken to attend patient conference to disseminate findings and impact of PPI in this research)
Type of cost: NHS
Please describe fully: Subsistence (vouchers for time taken to attend patient conference to disseminate findings and impact of PPI in this research)

Justification of Cost: Year 3: Cost to cover time for attending The Swallows conference (2 days) £150/day per INVOLVE guidance for 2 days, 5 PPI members = £1500.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1500	1500

Description: Subsistence (vouchers for time taken to attend stakeholder consultation)
Type of cost: NHS
Please describe fully: Subsistence (vouchers for time taken to attend stakeholder consultation)
Justification of Cost: Year 3: 2 PPI members to attend stakeholder consultation at £25/hr per INVOLVE guidance for 2 hrs = £100.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	100	100

Description: Subsistence - journey costs to attend patient conference
Type of cost: NHS
Please describe fully: Subsistence - journey costs to attend patient conference
Justification of Cost: Year 3: Travel costs to attend patient conference: Based on quotes from National Rail to previous locations, Sunderland and Torquay). Return rail fare from London for 5 people = £114.80 x 5 = £574 Taxi from train station to accommodation: £25 for 6-seater taxi per journey for outward and return =£50.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	624	624

Description: Subsistence - refreshments for attending patient conference
Type of cost: NHS
Please describe fully: Subsistence - refreshments for attending patient conference
Justification of Cost: Year 3: For attendance to conference/2 day away from home at £25/day per HMRC gov.uk guidance (https://www.gov.uk/government/publications/marine-licensing-fees/travel-and-subsistence-costs) for 5 PPI members = £250.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	250	250

Description: Subsistence - childcare costs to enable attendance of PPI members to patient conference
Type of cost: NHS
Please describe fully: Subsistence - childcare costs to enable attendance of PPI members to patient conference
Justification of Cost: None of my PPI members currently require this but as peoples circumstances may change, and if members leave and new people join during my PhD I am estimating childcare costs for approximately 50% of members (2 PPI members) requiring to attend the patient conference: Year 3: 2 days (48hrs) to allow attendance to The Swallows Conference for 2 PPI members = 48hrs at £15/hr per estimates from NIHR INVOLVE = £720.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	720	720

Description: Carer costs to enable attendance to patient conference
Type of cost: NHS
Please describe fully: Carer costs to enable attendance to patient conference
Justification of Cost: Carer costs based on £20/hr as mid-point for hourly rate and £200/day in NIHR Involve guidance of £200/day. Currently none of my PPI group require this but estimating costs if circumstances change at maximum 50% of PPI members requiring this to attend 2 day conference in Year 3 for 2 PPI members: 4 days at £200/day = £800.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	800	800

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	3015	1465	7232	11712

Other Direct Costs

Description: Production of animation for dissemination of findings
Type of cost: NHS
Please describe fully: Production of animation for dissemination of findings
Justification of Cost: To produce a 60 second digital animation for disseminating findings. Quote from UCL Creative Services – Animation and Motion graphics. Internal rate (£3000 exc vat).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	3600	3600

Description: Subsistence - vouchers for patients to renumerate time to attend interviews
Type of cost: NHS
Please describe fully: Subsistence - vouchers for patients to renumerate time to attend interviews
Justification of Cost: £25 voucher for participants will be provided per participant (estimated at n=10) as a thank you for their time. Each interview will last approximately 1 hr.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	250	0	250

Description: Subsistence - journey costs for participants to attend interviews and follow-up visit for cohort study
Type of cost: NHS
Please describe fully: Subsistence - journey costs for participants to attend interviews and follow-up visit for cohort study
Justification of Cost: My PPI group advised me that patients should be renumerated for travel costs to attend research activities. Costs are based on TfL quotes - London zone 1-6 day travel card (£15.20) to UCLH Macmillan Centre for the following activities: Year 2: Qualitative interviews n=10 patient study (maximum number). 10 x £15.20 = £152. Cohort study 1 month follow up outcome measures: for ease of participants, PG-SGA and weight will be collected at home, scales provided. PG-SGA does include a brief physical exam but this could be conducted virtually over video consultation for patients receiving follow up a local unit within the centralised pathway. In the event that patients cannot have virtual review, will cost in travel to UCLH MCC to collect these measures as an estimate of 50% of the group (45 x £15.20 = £684).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	836	0	836

Description: Subsistence - childcare for patients attending interviews
Type of cost: NHS
Please describe fully: Subsistence - childcare for patients attending interviews
Justification of Cost: Childcare for participants - maximum costed for 3hrs estimated on 50% of participants requiring this (1hr interview + 1hr travel each way) at £15/hr for 5 participants = 15hrs at £15/hr = £225.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	225	0	225

Description: Subsistence - carer costs to attend patient interviews
Type of cost: NHS
Please describe fully: Subsistence - carer costs to attend patient interviews
Justification of Cost: Carer costs for participants taking part in qualitative interviews - maximum costed for 3hrs estimated on 50% of participants requiring this (1hr interview + 1hr travel each way) at £20/hr (based on taking mid-point from NIHR Involve recommendations that range from £7.50-£40/hr) 5 participants x 3hrs = 15hrs at £20/hr = £300.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	300	0	300

Description: Subsistence - refreshments for stakeholder consultation
Type of cost: NHS
Please describe fully: Subsistence - refreshments for stakeholder consultation
Justification of Cost: Refreshments for stakeholder consultation meeting. £10/head per 6 staff members and 2 members of research team. (The 2 PPI members are costed into the PPI section).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	80	80

Description: Transcription
Type of cost: NHS
Please describe fully: Transcription
Justification of Cost: Transcription costs for qualitative interview transcripts. Quote from professional transcription service (www.waywith words.com) for interviews (costed 97p/minute for 1680 minutes based on maximum 28 interviews x 1hrs each).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1630	0	1630

Description: Interpreters for patient interviews
Type of cost: NHS
Please describe fully: Interpreters for patient interviews
Justification of Cost: Quotes from UCLH language line service: Face-Face interpretation = £23/hr. Year 2: Patient interviews: I will need to cost for the potential for interpretation requirements for the patient qualitative interviews. All staff have an English language requirement in their positions therefore not required for staff interviews. Local audit data found 14% of patients with HNC attending UCLH require interpretation services. Based on 14% of 10 interviews = 1-2 people. 2x 1 hour interview =£46 + 2x 30minutes to discuss study, translate information in the participant information sheet and consent patients =£23. Total = £23 + £46 = £69.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	69	0	69

Description: Language translation of study materials and outcome measures
Type of cost: NHS
Please describe fully: Language translation of study materials and outcome measures
Justification of Cost: A local audit indicated that 14% of patients attending UCLH required interpretation services/language barriers. Quotes have been obtained from UCLH language line services for the top 2 most common languages (Bengali and Turkish). Year 2/3: I have costed for translation of study materials that will be required by patients at home and without access to on-site interpreters. This includes: 1) Patient diet diary template (quoted at £138/language) 2) Completed diary (£597/language).14% of 90 = 13 documents = 1794+7761 = £9555. The PGSGA comes in a variety of languages and patient completed question are mainly tick boxes. 2 patient interview transcripts = £597/language = £1194.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	10749	0	10749

Description: Survey dissemination fee
Type of cost: NHS
Please describe fully: Survey dissemination fee
Justification of Cost: To disseminate survey to professional groups. No cost incurred – I have reached out to multiple professional bodies (BDA, BAHNO, BAHNON) all who have advised that they will be able to disseminate my survey free of charge.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Qualtrics software
Type of cost: NHS
Please describe fully: Qualtrics software
Justification of Cost: Healthcare professional online survey software is free of charge and recommended by UCL Information Services for staff/students developing a survey.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Covidence software
Type of cost: NHS

Please describe fully: Covidence software
Justification of Cost: Year 1: Software for managing scoping review data. To manage my scoping review searches and data extraction. $240 (£187) for 1 review with >1 user.

	Year 1	Year 2	Year 3	Total Cost (£)
	187	0	0	187

Description: Nutritics software
Type of cost: NHS
Please describe fully: Nutritics software
Justification of Cost: Nutritional analysis software for nutritional analysis of diet diaries. Website quotes £19/month for graduate license billed annually (2 years at £19/month = £456 exc tax). Professor Nathan Davies (Head of UCL Nutrition Department) has agreed to provide me with one of their teaching licenses free of charge.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Interpreters for participation in cohort study
Type of cost: NHS
Please describe fully: Interpreters for participation in cohort study
Justification of Cost: Quotes from UCLH language line service: Face-Face interpretation = £23/hr. Year 2: Local audit data found 14% of patients with HNC attending UCLH require interpretation services. 14% of 90 = 13x 30minutes to discuss study and consent patients =£149.50.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	0	150

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	187	14209	3680	18076

Training And Development

Type of cost: NHS
Description: Applicant PhD Tuition Fees
Item Description: Doctoral student fees
Justification of Cost: UCL Division of Surgery and Intervention Science PhD student fees (£6035/year) from 2024/25 for UK students: https://www.ucl.ac.uk/prospective-students/graduate/research-degrees/division-surgery-and-interventional-science-mphil-phd.

	Year 1	Year 2	Year 3	Total Cost (£)
	6035	6035	6035	18105

Type of cost: NHS
Description: Short Courses
Item Description: JBI scoping review course
Justification of Cost: Course fees (external delegate) for University of Nottingham JBI scoping review (online).

	Year 1	Year 2	Year 3	Total Cost (£)
	199	0	0	199

Type of cost: NHS
Description: Short Courses
Item Description: Qualitative research methods in health
Justification of Cost: Course fees for UCL qualitative research methods in health course (online, 10 half days).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1350	0	1350

Type of cost: NHS
Description: Short Courses

Item Description: Rapid Research Evaluation and Appraisal Bootcamp Training
Justification of Cost: Course fees for UCL rapid qualitative research course (face to face).

	Year 1	Year 2	Year 3	Total Cost (£)
	1250	0	0	1250

Type of cost: NHS
Description: Short Courses
Item Description: Ethnography for Healthcare Improvement Sumer School, University of Leicester.
Justification of Cost: Course fees for University of Leicester Summer School 2 days (12hrs).

	Year 1	Year 2	Year 3	Total Cost (£)
	650	0	0	650

Type of cost: NHS
Description: Short Courses
Item Description: Introduction to NVivo for qualitative and mixed methods analysis
Justification of Cost: Course fees (£255 student fee) for University of Surrey (2 days) in person course.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	255	0	255

Type of cost: NHS
Description: Short Courses
Item Description: Writing and Publishing Qualitative Research
Justification of Cost: Course fees for UCL Writing and Publishing Qualitative Research short course.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	75	75

Type of cost: NHS
Description: Short Courses
Item Description: Implementation Science Masterclass
Justification of Cost: Course fees for King's College London 2 day short course organised by NIHR ARC South London Non-KCL student/NHS fee = £600.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	600	0	600

Type of cost: NHS
Description: Short Courses
Item Description: Developing and Evaluating Complex interventions
Justification of Cost: Course fees for University of Glasgow course (5x2.5hrs online sessions). Student fee/session = £30. Total = £30x5 = £150.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	0	150

Type of cost: NHS
Description: Short Courses
Item Description: Rosalind Franklin Programme
Justification of Cost: Course fees £1200 (fully inclusive and all delivered online over 9 months).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1200	1200

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	8134	8390	7310	23834

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	97386	113777	112852	324015

Indirect costs

HEI Indirect costs
No costs have been added

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	0	0	0	0

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	4465	0	4465
Total Higher Education Institution Costs (100%)s	0	0	0
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	319550	0	319550
Total Research Costs Requested From Funder	324015	0	324015

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**81897**	**85677**	**94126**	**261700**
Travel-subsistence-and-conference-fees	**1918**	**2227**	**471**	**4616**
Equipment	**2226**	**1356**	**24**	**3606**
Consumables	**9**	**453**	**9**	**471**
Patient-and-Public-Involvement	**3015**	**1465**	**7232**	**11712**
Other-Direct-Costs	**187**	**14209**	**3680**	**18076**
Training And Development	**8134**	**8390**	**7310**	**23834**

HEI-Indirect-costs	**0**	**0**	**0**	**0**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**97386**	**113777**	**112852**	**324015**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
Eligibility checks	0	364	364	728.0
Informed consent	0	1467	1613	3080.0

Justification:
Eligibility checks: Eligibility checks for cohort study (Nb: patients for interviews/observations will be recruited from the cohort study). Eligibility will be predominantly completed by myself with support from clinical staff who attend the weekly 3hr HNC multidisciplinary meeting where new patients are discussed and would be considered for eligibility for inclusion. Total = £727.50 spread across Years 2 and 3. **Informed consent:** Costs to cover time taken to take informed consent from patients for cohort study (£2182.50 for 90 participants years 2/3), interviews/observations (£751.75 years 2/3) for 10 patients, 18 staff and 3 patient observation cases, and stakeholder consultation (£145.50 year 3) for 6 staff members, (PPI members will be there in advisory roles). Rates taken from 2022 SoECAT standard rates (15 minutes/ £24.25 per participant).

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Support Costs Total	0	1831	1977	3808

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
No

Usual Treatment Costs

No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	3808
NHS Excess Treatment costs/savings	0

TOTAL NHS SUPPORT AND TREATMENT COSTS	3808

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	324015
Total NHS (Support and Excess Treatment) Costs	3808
Total Funding Required	327823

10. Uploads

(References - References.docx) is included as an appendix within this file.

(Figures / Tables - Supplementary figures(2).docx) is included as an appendix within this file.

(Research Timetable - Research timetable.docx) is included as an appendix within this file.

(SoECAT Template - Funder Export_334541_SoECAT__SUSTENANCE_ Development of a post-operative feeding toolkit i.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Professor
Forename(s)	Suneetha Ramani
Surname	Moonesinghe
Position	Professor and Honorary Consultant

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Dr
Forename(s)	Roganie
Surname	Govender
Position	Consultant Speech and Language Therapist

Supervisor Participant 2

Title	Dr
Forename(s)	Cecilia
Surname	Vindrola
Position	

Practice Supervisor(s)

Practice Support Participant 1

Title	Professor
Forename(s)	Peter
Surname	Brennan
Position	Consultant Surgeon

Practice Support Participant 2

Title	Dr
Forename(s)	Alastair
Surname	Duncan
Position	Consultant Dietitian

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Professor
Forename(s)	Suneetha Ramani
Surname	Moonesinghe
Position	Professor and Honorary Consultant

Head of Department or Senior Manager 2

Title	Dr
Forename(s)	fiona
Surname	yung
Position	Divisional Manager

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mr
Forename(s)	Joe
Surname	Mwanza
Position	Head of Finance

12. Acknowledgement, Review and Submit

Conflict checks:
Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

I declare no conflicts of interest.

Agreement to Terms and Conditions:

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. References.docx
2. Supplementary figures(2).docx
3. Research timetable.docx
4. Funder Export_334541_SoECAT__SUSTENANCE_ Development of a post-operative feeding toolkit i.xlsx

Appendix #1: References.docx

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Appendix #2: Supplementary figures(2).docx

Figure 1: The pathway for how patients currently transition from tube to oral feeding after undergoing flap surgery for head and neck cancer.

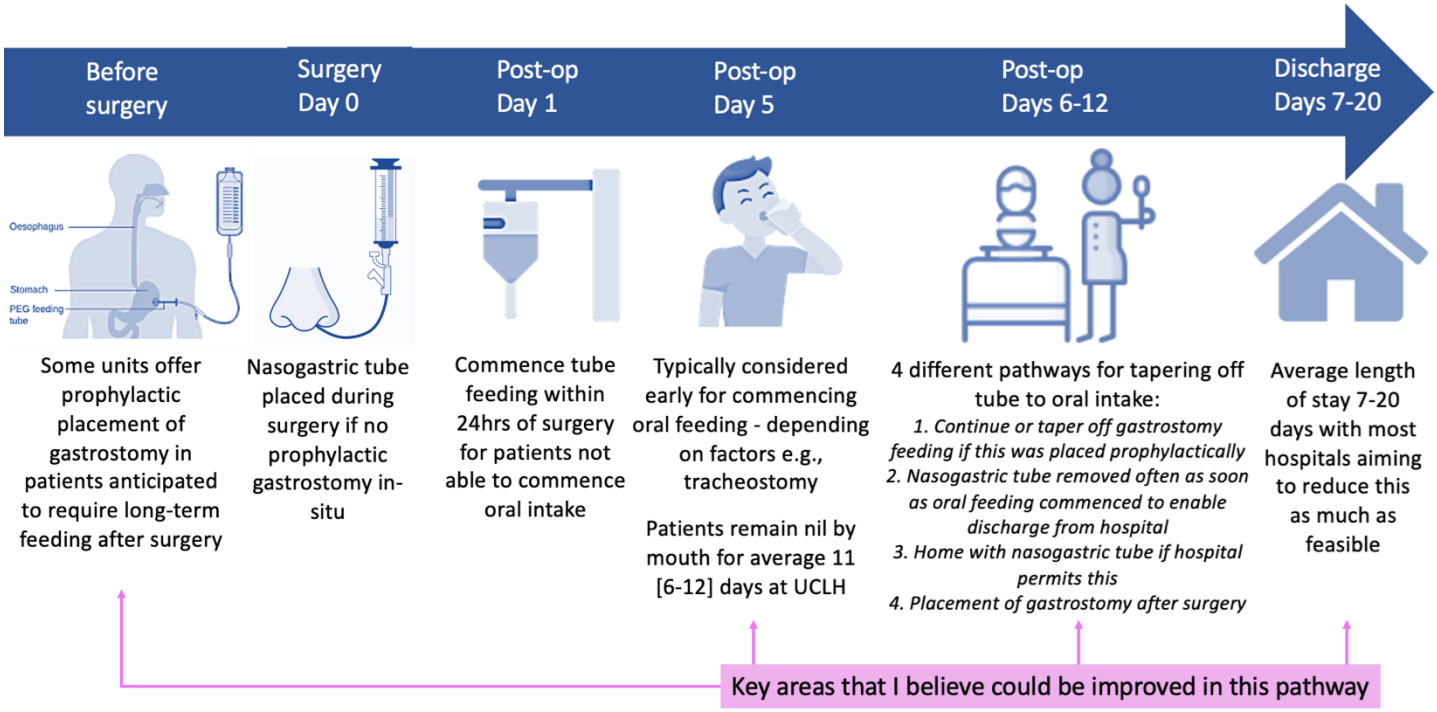
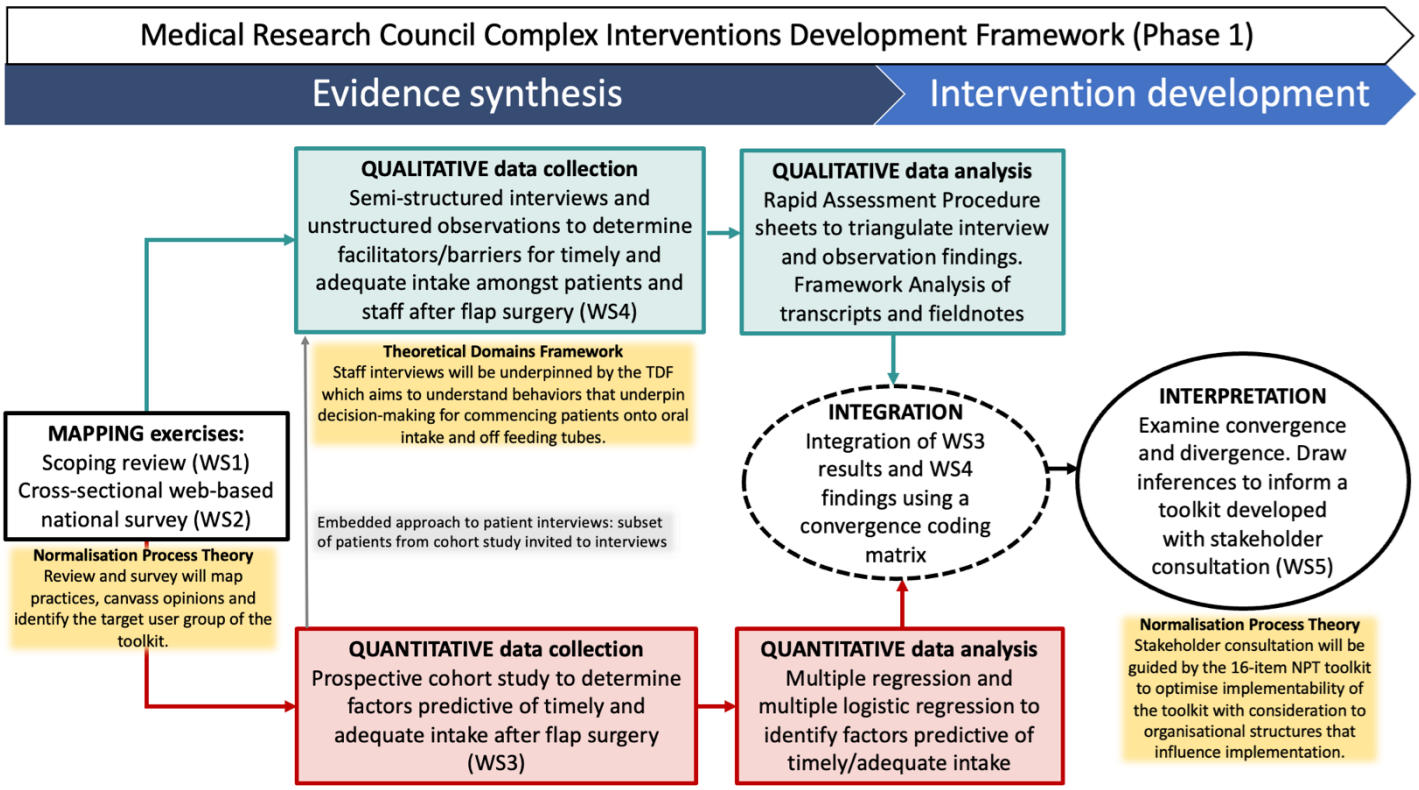


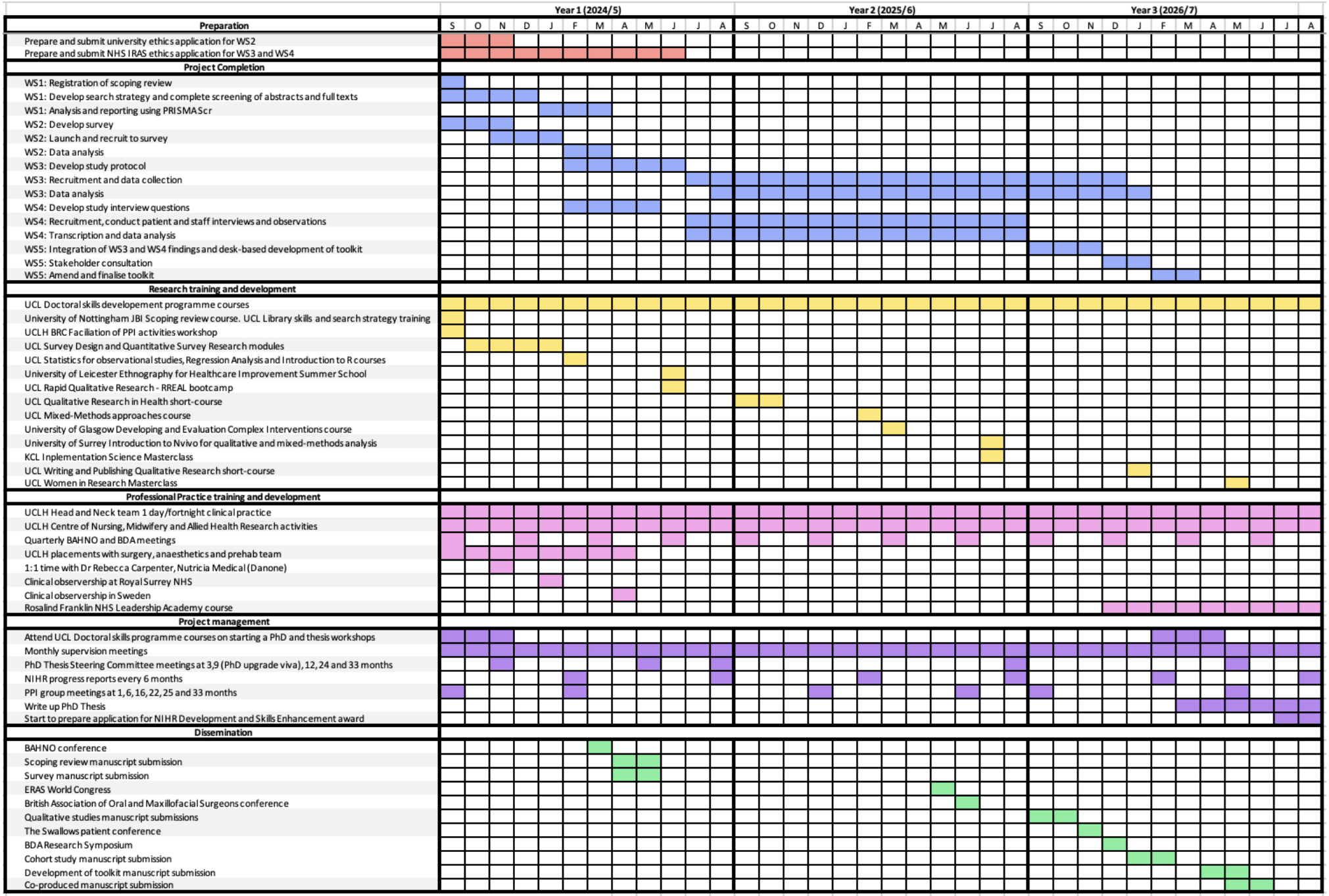
Figure 2: Convergent-parallel mixed-methods study design with application of underpinning theories and frameworks.



This mixed-methods study will address post-operative feeding practices and nutritional intake in patients with head and neck cancer transitioning from tube to oral feeding after undergoing flap surgery. A convergent-parallel mixed-methods design will be used, and it is a type of design in which qualitative and quantitative data are collected in parallel, analysed separately and then merged. In this study, a prospective cohort study will be used to identify factors that may be predictive of timely and adequate oral intake in patients with head and neck cancer after undergoing flap surgery. Semi-structured interviews and unstructured observations will explore barriers and facilitators amongst patients and healthcare professionals for transitioning from tube to adequate oral intake. The reason for collecting both quantitative and qualitative data is to obtain different, multiple perspectives that contribute to achieving timely and adequate intake to build a comprehensive understanding that can inform the development of a toolkit to improve post-operative feeding practices and nutritional intake in patients with head and neck cancer after undergoing flap surgery.

Appendix #3: Research timetable.docx

Research timetable (Based on 1.0WTE (20% time clinical) over 3 years commencing September 2024 (48 weeks/year accounting annual leave/bank holidays)



Appendix #4: Funder Export_334541_SoECAT__SUSTENANCE_ Development of a post-operative feeding toolkit i.xlsx